

ENGINEERING CHEMISTRY LAB

1. Engineering Chemistry is the study of the chemistry of materials used in engineering applications.
2. It involves the study of the physical, chemical and biochemical properties of materials used in engineering.
3. The engineering chemistry lab is used to study the various properties of materials used in engineering processes.
4. In the lab, experiments are conducted to determine the physical and chemical properties of materials such as metals, plastics, ceramics and composites.
5. The lab also studies the interaction of materials with different environments, such as with different temperatures, pH levels, and chemical or biological agents.
6. The lab also studies the corrosion behavior of materials and the effect of corrosion on materials.
7. The lab also studies the behavior of materials in extreme environments, such as high temperatures, high pressures, and in the presence of radiation.
8. The lab also studies the behavior of materials in different manufacturing processes, such as casting, welding, and machining.
9. The lab also studies the behavior of materials in different service environments, such as in automotive, aerospace, and marine applications.
10. The lab also studies the behavior of materials in different environmental conditions, such as in the presence of humidity, dust, and other contaminants.

Course Objectives:

- Understand the fundamental principles of computer science
- Develop an understanding of the core concepts of programming
- Learn to design, implement and evaluate algorithms
- Develop an awareness of the impact of computing on society
- Develop problem-solving skills and the ability to use computers to solve problems
- Understand the ethical, legal, and social implications of computing

1. To enable the students to understand about the fundamental concepts of analytical instruments

1. Analytical instruments are used to measure physical and chemical properties in order to gain insight into the composition and characteristics of a sample.
2. These instruments can be used in a variety of fields, such as food and beverage testing, environmental monitoring, pharmaceutical analysis, and more.
3. Common analytical instruments include spectroscopy, chromatography,

and mass spectrometry.

4. Spectroscopy is a technique that uses light or other forms of electromagnetic radiation to measure the composition of a sample.
5. Chromatography is a separation technique that uses a stationary phase and a mobile phase to separate components of a sample.
6. Mass spectrometry is a technique used to identify and quantify the chemical components of a sample.
7. Analytical instruments can be used to measure physical properties such as pH, viscosity, and refractive index, as well as chemical properties such as concentration, purity, and stability.
8. These instruments can also be used to analyze complex mixtures and detect trace amounts of substances.
9. The accuracy and precision of the measurements taken with analytical instruments depend on the quality of the instrument and the skill of the operator.

2. To enable the students to understand about the analysis of chloride content, hardness, alkalinity of water:

1. Chloride content can be determined by titration with silver nitrate, using a spectrophotometer to measure the intensity of the silver chloride color.
2. Hardness of water can be determined by EDTA titration, using a colorimeter to measure the intensity of the EDTA-metal ion complex.
3. Alkalinity of water can be measured by titration with hydrochloric acid, using a pH meter to measure the pH of the solution.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of a liquid.

- **pH** is a measure of the acidity or basicity of a liquid. It is measured on a scale from 0 to 14, with 7 being neutral. A pH below 7 is acidic, and a pH above 7 is basic.
- **Surface tension** is a measure of the cohesion of the molecules of a liquid. It is measured in units of force per unit length, such as dynes/cm.
- **Viscosity** is a measure of the resistance of a liquid to flow. It is measured in units of force times distance, such as poise or centipoise.

4. To enable the students to understand about the preparation of different resins

1. Resin is a natural or synthetic organic compound that is typically viscous, transparent, and soluble in organic solvents.
2. Resins are typically formed through the polymerization of monomers and can be either thermoplastic or thermosetting.
3. Thermosetting resins are further classified into two types: condensation and addition.

4. Condensation resins are formed through the reaction of two or more molecules with the elimination of a small molecule like water or alcohol. Examples include phenolic and epoxy resins.
5. Addition resins are formed through the reaction of two or more molecules without the elimination of a small molecule. Examples include polyurethanes and acrylics.
6. Resins can be prepared from a variety of sources such as plants, animals, minerals, and synthetic materials.
7. Synthetic resins are produced in a laboratory by combining two or more molecules. Examples include polyvinyl chloride (PVC) and polyethylene terephthalate (PET).
8. Resins can be used in a variety of applications such as adhesives, coatings, and composites.

5. To enable the students to understand about the synthesis of organic compounds such as adipic acid and paracetamol by conventional and green route.

1. Organic compounds are molecules that contain carbon and hydrogen atoms, and are found in living organisms.
2. Adipic acid is a dicarboxylic acid that is used in the production of nylon, polyurethanes and other synthetic materials.
3. Paracetamol is a widely used analgesic and antipyretic drug.
4. Conventional synthesis of organic compounds such as adipic acid and paracetamol involves the use of toxic chemicals and solvents.
5. Green synthesis of organic compounds is a less hazardous and more environmentally friendly approach that uses natural catalysts such as enzymes and microorganisms.
6. Green synthesis of adipic acid and paracetamol has been successfully demonstrated using biocatalysts such as lipases and oxidases.
7. Green synthesis of organic compounds is an important area of research, and has potential applications in the pharmaceutical and chemical industries.

List of Experiments

1. Boyle's Law Experiment: This experiment was conducted by Robert Boyle in 1662 and is used to illustrate the relationship between pressure and volume of a gas.
2. Faraday's Law of Induction Experiment: This experiment was conducted by Michael Faraday in 1831 and is used to demonstrate the production of an electric current in a closed circuit when a magnetic field is applied.
3. Ohm's Law Experiment: This experiment was conducted by Georg Ohm in 1827 and is used to illustrate the relationship between current, voltage, and resistance in an electrical circuit.
4. Avogadro's Law Experiment: This experiment was conducted by Amedeo

Avogadro in 1811 and is used to illustrate the relationship between the volume of a gas and its amount.

5. Charles' Law Experiment: This experiment was conducted by Jacques Charles in 1787 and is used to illustrate the relationship between temperature and volume of a gas.
6. Graham's Law of Diffusion Experiment: This experiment was conducted by Thomas Graham in 1851 and is used to illustrate the relationship between the rate of diffusion of two gases and their molecular weights.
7. Dalton's Law of Partial Pressure Experiment: This experiment was conducted by John Dalton in 1801 and is used to illustrate the relationship between the total pressure of a gas and the partial pressure of its components.

1. Calibration of Analytical Equipment and Apparatus

1. Calibration is the process of comparing the performance of a device to a known standard and adjusting it accordingly.
2. It is important to ensure that the analytical equipment and apparatus used in laboratories are properly calibrated to achieve accurate results.
3. The calibration process includes setting up the equipment, verifying its performance, and adjusting the equipment to meet a predetermined standard.
4. The calibration of analytical equipment and apparatus can be done manually or with the help of automated systems.
5. It is important to ensure that the calibration process is done regularly and accurately to ensure the accuracy of the results.
6. The calibration process should be documented and the results should be compared to the expected results.
7. The calibration process should be done in accordance with the manufacturer's instructions and the applicable standards.
8. The calibration process should be documented and the results should be reviewed regularly.
9. The calibration process should be done in a controlled environment to ensure the accuracy of the results.
10. The calibration process should be done in accordance with the applicable safety regulations.

2. Determination of Hardness of Water Sample by EDTA Method

1. Hardness of water can be determined by titration with a chelating agent such as EDTA.
2. EDTA reacts with metal ions in water to form a complex that can be titrated.
3. The endpoint of the titration is determined by an indicator such as Eriochrome Black T.
4. The amount of EDTA used in the titration is proportional to the concen-

tration of metal ions in the sample.

5. The total hardness of a sample can be calculated by multiplying the molarity of EDTA used in the titration by a factor of 5.
6. The EDTA method for determining water hardness is accurate and reliable, but it requires a skilled operator and specialized equipment.

3. Determination of Alkalinity of water sample

- Alkalinity is a measure of the ability of water to neutralize acids and is expressed in terms of the equivalent concentration of calcium carbonate.
- Alkalinity can be determined by titration with a strong acid, such as hydrochloric acid or sulfuric acid, using a pH meter or color indicator.
- The end point of the titration is indicated by either a change in the pH of the solution or the appearance of a color indicator.
- The alkalinity of the sample can then be calculated using the volume of acid added and the titrant concentration.
- The alkalinity of natural water samples can be determined through the use of a carbonate-bicarbonate titration.
- This method involves adding a known amount of acid to the sample and titrating until the pH of the sample reaches a certain level.
- The alkalinity can then be calculated from the volume of acid added and the titrant concentration.

4. Determination of pH by titrimetric method

- Titrimetric method is a process of determining the concentration of a substance in a solution by adding a standard solution of known concentration.
- The process involves adding a standard solution of known concentration to the solution containing the unknown concentration of the substance.
- The pH of the solution is then determined by measuring the amount of standard solution required to reach the equivalence point.
- The equivalence point is the point where the number of moles of the titrant (standard solution) is equal to the number of moles of the analyte (unknown solution).
- The pH of the solution can be calculated by using the formula, $\text{pH} = -\log[\text{H}^+]$.
- The titrimetric method is an accurate and reliable method for determining the pH of a solution. It can be used for both strong and weak acids and bases.
- The method is also useful for determining the pH of complex solutions such as buffers, where the pH can be affected by the presence of other ions.

5. Determination of surface tension of given liquid

1. Surface tension is the property of a liquid that causes it to resist an external force.

2. Surface tension is a result of the cohesive forces between the molecules of the liquid.
3. Surface tension can be measured using a variety of methods, such as the Wilhelmy plate method or the capillary rise method.
4. In the Wilhelmy plate method, a plate is placed on the surface of the liquid and the force required to pull it up is measured.
5. In the capillary rise method, the liquid is drawn up a capillary tube and the height to which it rises is measured.
6. The surface tension of a liquid can be used to calculate its surface energy.

6. Determination of Viscosity of a given liquid by viscometer

1. Viscosity is a measure of the resistance of a liquid to flow. It is typically measured with a viscometer.
2. A viscometer is a device that measures the viscosity of a liquid. It consists of a tube with a rotating paddle inside. As the paddle rotates, the viscosity of the liquid is measured.
3. To measure the viscosity of a given liquid, the liquid is placed in the viscometer and the paddle is rotated. The viscosity is then calculated based on the speed at which the paddle rotates.
4. Viscosity is an important property of a liquid, as it affects the flow characteristics of the liquid. It is used to determine the flow rate of a liquid, and is also used to determine the stability of a liquid.
5. Viscosity can be measured using a variety of methods, including rotational viscometers, cone and plate viscometers, and falling ball viscometers. Each method has its own advantages and disadvantages, and the method chosen should be based on the application.
6. The viscosity of a liquid can be affected by temperature, pressure, and the presence of other substances. Therefore, it is important to take into account these factors when measuring the viscosity of a liquid.

7. Determination of the Strength of Ferrous Ammonium Sulfate Using External Indicator

1. Ferrous ammonium sulfate is an inorganic compound that is composed of iron, ammonium, and sulfate ions.
2. This compound is commonly used in analytical chemistry to determine the strength of a solution, and is often used in titration experiments.
3. An external indicator is a chemical compound that can be used to determine the end point of a titration.
4. The most commonly used external indicator for ferrous ammonium sulfate is thymol blue.
5. Thymol blue changes color when exposed to ferrous ammonium sulfate, and this color change indicates the end point of the titration.

6. The strength of ferrous ammonium sulfate can be determined by adding a known amount of thymol blue to the solution and observing the color change.
7. Once the color change has been observed, the strength of the ferrous ammonium sulfate can be determined by comparing the color change to a standard chart.

8. Determination of the Strength of Potassium Dichromate Using Internal Indicator

1. Potassium dichromate is an inorganic compound, with the chemical formula $K_2Cr_2O_7$. It is a strong oxidizing agent and is used in a variety of industrial and laboratory applications.
2. An internal indicator can be used to determine the strength of a potassium dichromate solution. This indicator is a compound which changes colour at a certain pH.
3. To determine the strength of a potassium dichromate solution, a sample of the solution is taken and a few drops of the internal indicator are added. The colour change of the indicator will indicate the strength of the solution.
4. The strength of a potassium dichromate solution can also be determined by titration. In this method, a known volume of the potassium dichromate solution is titrated with a standard solution of a base. The volume of the base required to neutralize the acid is an indication of the strength of the potassium dichromate solution.
5. The strength of a potassium dichromate solution can also be determined by spectrophotometry. In this method, the absorbance of the solution is measured at a certain wavelength. The absorbance of the solution is an indication of the strength of the solution.

9. Determination of available chlorine in bleaching powder

1. Bleaching powder is a white powder made from calcium chloride and sodium hypochlorite.
2. It is used as a disinfectant and bleaching agent in laundry and other household cleaning products.
3. The available chlorine in bleaching powder can be determined by titration with sodium thiosulfate solution.
4. A known amount of bleaching powder is dissolved in water and titrated with sodium thiosulfate solution.
5. The number of moles of sodium thiosulfate required to neutralize the bleaching powder is equal to the number of moles of available chlorine in the sample.

6. The concentration of available chlorine in the sample can be calculated by dividing the number of moles of available chlorine by the volume of the sample.

10. Determination of Chloride Content in Water Sample

1. Chloride content in water samples can be determined using a variety of methods, such as titration, gravimetric analysis, and colorimetric analysis.
2. Titration is a quantitative analytical method used to determine the concentration of a known reactant in a solution. It involves adding a known amount of a reagent to a solution until the reaction is complete.
3. Gravimetric analysis is a method of analyzing the mass of an unknown substance in a sample. This method involves weighing a sample before and after a reaction to determine the mass of the unknown substance.
4. Colorimetric analysis is a method of analyzing the concentration of a substance in a sample by measuring the intensity of light absorbed or emitted by the sample.
5. Chloride content in water samples can also be determined using ion-selective electrodes. This method involves measuring the electrical potential of the sample to determine the concentration of chloride ions.

11. Preparation of Phenol formaldehyde (PF) Resin

1. Phenol formaldehyde (PF) resin is a thermosetting synthetic polymer produced from the reaction of phenol and formaldehyde.
2. The reaction of phenol and formaldehyde is known as a condensation reaction, which occurs when two molecules combine to form a larger molecule, with the release of a small molecule such as water or methanol.
3. The reaction is typically carried out in the presence of an acid catalyst, such as sulfuric acid, to speed up the reaction.
4. The reaction is exothermic, meaning that it produces heat.
5. The resulting resin is a solid, insoluble material with a high molecular weight.
6. The properties of the resin depend on the ratio of phenol to formaldehyde used in the reaction, as well as the type of acid catalyst and the reaction conditions.
7. The resin is often used as an adhesive, coating or molding material in a variety of industrial applications.

12. Preparation of Urea formaldehyde (UF) Resin

1. Urea formaldehyde (UF) resin is a synthetic polymer used in many applications such as adhesives, coatings, and composites.
2. It is formed by the condensation reaction of urea and formaldehyde.
3. Urea formaldehyde resin is produced by reacting urea with formaldehyde under basic (alkaline) conditions.

4. The reaction is usually carried out in an aqueous solution of formaldehyde, which is typically stabilized with a base such as sodium hydroxide or ammonium hydroxide.
5. The reaction is exothermic, and the temperature of the reaction must be carefully controlled to ensure that the product is not overheated.
6. The reaction is typically carried out at a temperature of 70-80°C, and the reaction time is usually between 3-4 hours.
7. The reaction is typically carried out in the presence of an acid catalyst, such as sulfuric acid or hydrochloric acid, to increase the rate of reaction.
8. The reaction may also be carried out in the presence of a surfactant, such as polyvinyl alcohol, to improve the solubility of the product.
9. The reaction is typically carried out under pressure, and the pressure is typically maintained at between 10-15 bar.
10. The reaction is typically carried out in the presence of a cross-linking agent, such as hexamethylene diisocyanate, to increase the molecular weight of the product.
11. The reaction is typically carried out in the presence of a stabilizer, such as an amine or a phenol, to reduce the rate of hydrolysis of the product.
12. The product of the reaction is a solid, insoluble polymer with a high molecular weight.

13. Preparation of Adipic acid / Paracetamol

- Adipic acid is a white, crystalline solid that is soluble in water. It is used in the production of nylon and is also used as a food additive.
- Paracetamol is a white, crystalline solid that is also soluble in water. It is used as an analgesic and antipyretic and is also used as a food additive.
- Adipic acid can be prepared by the oxidation of cyclohexanol, using nitric acid as the oxidizing agent.
- Paracetamol can be prepared by the reaction of p-aminophenol with acetic anhydride, followed by neutralization with sodium hydroxide.

14. Determination of Cell Conductance of a Solution

1. Cell conductance is a measure of the ability of a solution to conduct electricity.
2. It is expressed as the ratio of the electrical current passing through the solution to the potential difference across it.
3. Cell conductance is affected by the concentration of ions in a solution, as well as the temperature.
4. The cell conductance of a solution can be determined using a conductivity meter.
5. This instrument is used to measure the electrical resistance of a solution by passing a known current through it.
6. The cell conductance is then calculated from the measured resistance and the applied voltage.

7. The cell conductance of a solution can also be determined by measuring the current and voltage across a known resistance in the solution.
8. The cell conductance is then calculated from the measured current and voltage.

15. Determination of Rate Constant of Hydrolysis of Esters

1. Hydrolysis of esters is a chemical reaction where an ester molecule is broken down into an alcohol and an acid.
2. The rate constant of hydrolysis of esters is a measure of how quickly the reaction occurs. It is typically expressed in terms of the number of moles of ester that are broken down per second.
3. The rate constant can be determined experimentally by measuring the concentration of ester molecules over time.
4. The rate constant is affected by several factors, including the temperature and the presence of a catalyst.
5. The rate constant can be used to calculate the rate of hydrolysis of an ester at different temperatures and in the presence of a catalyst.
6. The rate constant can also be used to determine the half-life of an ester, which is the amount of time it takes for half of the ester molecules to be broken down.

16. Element Detection and Identification of Functional Groups in Organic Compounds

1. Organic compounds are composed of a variety of elements, including hydrogen, carbon, nitrogen, oxygen, and other trace elements.
2. Functional groups are specific arrangements of atoms within a molecule that determine the chemical and physical properties of the compound.
3. Element detection is the process of determining the elements present in a given compound. This is typically done using spectroscopic techniques, such as infrared spectroscopy (IR), nuclear magnetic resonance (NMR), and mass spectrometry (MS).
4. Identification of functional groups is the process of determining the specific arrangement of atoms that form a particular functional group. This can be done by examining the chemical structure of the compound, or by analyzing the spectroscopic data.
5. IR spectroscopy is a powerful tool for detecting and identifying functional groups in organic compounds. It is based on the fact that different functional groups absorb different frequencies of infrared light, allowing them to be distinguished from one another.
6. NMR spectroscopy is another useful tool for detecting and identifying functional groups in organic compounds. It is based on the fact that

different nuclei (e.g., hydrogen and carbon) absorb different frequencies of radio waves, allowing them to be distinguished from one another.

7. Mass spectrometry is a powerful tool for detecting and identifying organic compounds. It is based on the fact that different molecules have different masses, allowing them to be distinguished from one another.
8. Other techniques, such as X-ray crystallography and chromatography, can also be used to detect and identify functional groups in organic compounds.

Topic: Experiments

1. Conduct an experiment to measure the boiling point of water.
2. Perform an experiment to determine the density of an unknown liquid.
3. Perform an experiment to measure the rate of photosynthesis in plants.
4. Conduct an experiment to measure the rate of respiration in animals.
5. Perform an experiment to measure the rate of diffusion in a solution.
6. Conduct an experiment to measure the conductivity of a solution.
7. Perform an experiment to measure the rate of evaporation of a liquid.
8. Conduct an experiment to measure the viscosity of a liquid.
9. Perform an experiment to measure the rate of fermentation in yeast.
10. Conduct an experiment to measure the rate of oxidation in a reaction.

Course Outcomes:

- Develop an understanding of the fundamental principles of computer science and engineering, including algorithms, data structures, operating systems, and computer architecture.
- Learn to design, implement, and evaluate computer-based systems, processes, components, and programs.
- Develop an understanding of the basic principles of software engineering, including software design, software development, and software testing.
- Acquire the skills to use modern software engineering tools and techniques.
- Learn to apply mathematics, science, and engineering to solve problems in computer science and engineering.
- Learn to identify, analyze, and solve complex computing problems.
- Develop an understanding of the ethical and social implications of computing and its applications.

Upon completion of the course the student should be able to:

- Identify and explain key concepts related to the course
- Apply key concepts to real-world scenarios
- Analyze and synthesize data to draw meaningful conclusions
- Develop and evaluate strategies for problem-solving
- Critically evaluate and articulate the implications of the course material

- Demonstrate an understanding of the ethical and legal considerations associated with the course material

Course Outcomes Bloom's

1. Understand the fundamental concepts of a course.
2. Analyze and interpret course material in order to draw conclusions.
3. Apply course material to solve problems.
4. Evaluate course material in order to identify strengths and weaknesses.
5. Create new solutions based on course material.

Level

1. Level is a term used to describe a particular stage in a process or development. It can be used to refer to a stage of education, a stage of development, or a stage in a game or other activity.
2. In education, levels are used to refer to the different stages of learning. For example, kindergarten is a level of education, as is elementary school, middle school, high school, college, and post-graduate studies.
3. In development, levels can refer to different stages of a process. For example, a company may have different levels of management, such as junior, mid-level, and senior.
4. In games and other activities, levels are used to refer to different stages of progress. For example, a video game may have different levels that the player must complete in order to progress to the next stage.
5. Levels can also be used to refer to different levels of difficulty. For example, a game may have different levels of difficulty, such as easy, medium, and hard.

CO-1 Get an Understanding of the Use of Different Analytical Instruments. K3

1. Analytical instruments are tools used to measure, analyze, and interpret physical properties of substances.
2. Common analytical instruments include spectrophotometers, chromatographs, mass spectrometers, microscopes, and refractometers.
3. Spectrophotometers measure the amount of light absorbed by a substance, which can be used to determine the concentration of a solution.
4. Chromatographs are used to separate and analyze mixtures of substances.
5. Mass spectrometers measure the mass-to-charge ratio of ions in a sample, which can be used to identify the components of a mixture.
6. Microscopes are used to observe and analyze small particles and structures.
7. Refractometers measure the refractive index of a substance, which can be used to determine the concentration of a solution.

CO-2 Measure the Molecular/System Properties Such as Surface Tension

1. Carbon dioxide (CO₂) is a compound composed of two oxygen atoms covalently bonded to a single carbon atom.
2. The physical properties of CO₂ are determined by its molecular structure.
3. At room temperature, CO₂ is a colorless, odorless gas.
4. The boiling point of CO₂ is -78.5°C (-109.3°F) and the melting point is -57.8°C (-72.8°F).
5. The density of CO₂ is 1.98 kg/m³ at 25°C (77°F).
6. The surface tension of CO₂ is 0.072 N/m.
7. The viscosity of CO₂ is 0.0018 Pa · s.
8. The refractive index of CO₂ is 1.00045.
9. The solubility of CO₂ in water is 0.8 g/L at 25°C (77°F).

Viscosity, Conductance of Solution, Chloride and Iron Content in the Water

- Viscosity is a measure of a fluid's resistance to flow. It is a measure of the internal friction of a fluid that is caused by the attraction between its molecules. The higher the viscosity, the slower the fluid flows.
- Conductance of a solution is a measure of its ability to conduct electrical current. It is proportional to the concentration of ions in the solution and is inversely proportional to the solution's resistance.
- Chloride is an anion found in many natural and synthetic compounds. It is present in seawater and is the most abundant anion in the oceans. It is also found in many minerals and soil.
- Iron is an essential nutrient for plants and animals. It is found in many foods and is an important component of hemoglobin, the molecule that carries oxygen in the blood. Iron is also found in many soils and is an important component of many industrial processes.

K3

- K3 is a mathematical field that studies algebraic varieties.
- Algebraic varieties are geometric objects defined by polynomial equations.
- Examples of algebraic varieties include circles, ellipses, hyperbolas, and parabolas.
- K3 surfaces are a special class of algebraic varieties that have been extensively studied.
- K3 surfaces have many interesting properties, such as being Calabi-Yau manifolds and having a rich moduli space.
- K3 surfaces are important in string theory and mirror symmetry, and have applications in other areas of mathematics and physics.

CO-3 Measure the hardness and alkalinity of the water. K3

1. Hardness of water is a measure of the amount of calcium and magnesium ions present in it.
2. Alkalinity of water is a measure of the amount of carbonate, bicarbonate, hydroxide, and other alkaline substances present in it.
3. Hardness of water is usually measured in parts per million (ppm) or milligrams per liter (mg/L).
4. Alkalinity of water is usually measured in milligrams per liter (mg/L).
5. Hardness of water can be determined by using a titration process with a calcium chloride solution.
6. Alkalinity of water can be determined by using a titration process with a sodium hydroxide solution.
7. High levels of hardness and alkalinity in water can cause scaling and staining of plumbing fixtures, and can also interfere with the effectiveness of detergents.
8. It is important to regularly test the hardness and alkalinity of water to ensure that it is safe for use.

CO-4 Know the fundamental concepts of the preparation of phenol

1. Phenol is an organic compound with the chemical formula C_6H_5OH . It is a colorless, volatile, flammable liquid with a characteristic odor.
2. Phenol can be prepared by several methods, including the Williamson ether synthesis, the Grignard reaction, and the Kolbe-Schmitt reaction.
3. The Williamson ether synthesis involves the reaction of an alkyl halide with an alcohol in the presence of a base, such as sodium hydroxide, to form an ether.
4. The Grignard reaction involves the reaction of an alkyl or aryl halide with a Grignard reagent (a magnesium halide) to form an alcohol.
5. The Kolbe-Schmitt reaction involves the reaction of an acid with a carbonate or bicarbonate to form an organic compound.
6. Phenol can also be prepared by the oxidation of benzene or other aromatic compounds.

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

- Formaldehyde is a colorless, flammable gas with a strong odor. It is used in a variety of industrial and consumer products, including adhesives, paints, and plastics. Urea formaldehyde resin is a thermosetting plastic made from a reaction between formaldehyde and urea. It is used in many products, such as furniture, flooring, and cabinetry.
- Adipic acid is a white, odorless, crystalline solid. It is used in the manufacture of nylon, polyurethanes, and other synthetic fibers.
- Paracetamol is an analgesic drug used to treat pain and fever. It is available in tablet and liquid form.

K3

1. K3 is a term used to refer to the three most important elements of knowledge management: knowledge creation, knowledge sharing, and knowledge application.
2. Knowledge creation is the process of generating new ideas and insights that can be used to improve an organization's operations. It involves research, experimentation, and analysis.
3. Knowledge sharing is the process of transferring knowledge from one person to another. It involves communication, collaboration, and networking.
4. Knowledge application is the process of using knowledge to solve problems and make decisions. It involves problem-solving, decision-making, and implementation.

CO-5 Estimate the rate constant of Reaction. K3

1. The rate constant of a reaction is a measure of how quickly the reaction takes place.
2. The rate constant is usually expressed in terms of the concentration of reactants over time.
3. The rate constant is usually denoted by the symbol 'K'.
4. The rate constant of a reaction is determined by the temperature, pressure and concentration of the reactants.
5. The rate constant of a reaction can be estimated by measuring the rate of the reaction at different temperatures and pressures.
6. The rate constant of a reaction can also be estimated by measuring the concentration of reactants over time.
7. The rate constant of a reaction can also be estimated by using the Arrhenius equation.
8. The Arrhenius equation can be used to calculate the rate constant of a reaction at any given temperature.

Topic:

- 1.
- 2.
- 3.

Please provide a specific topic for me to write about.

The Impact of Technology on Society

1. Technology has had a profound impact on the way people communicate, interact, and work.
2. Technology has enabled people to access information quickly and easily, and to connect with people all over the world.
3. Technology has also changed the way people do business, with businesses now relying heavily on digital technology to operate.
4. Technology has also changed the way people consume media, with streaming services providing access to a wide range of content.
5. Technology has also had a major impact on the environment, with the rise of renewable energy sources, electric vehicles, and other forms of green technology.
6. Technology has also had a major impact on the way people live, with smart home technology providing convenience and comfort.
7. Technology has also had a major impact on education, with online learning and virtual classrooms becoming more popular.
8. Technology has also had a major impact on healthcare, with the development of medical technology and the increasing use of telemedicine.
9. Technology has also had a major impact on the way people entertain themselves, with the rise of video games and streaming services providing access to a wealth of content.

Topic:

- 1.
- 2.
- 3.

Please provide me with more information about the topic you would like me to write about.

Topic:

- 1.
- 2.
- 3.
- 4.
- 5.

Please provide the topic for which you would like me to write a study material.

Topic: AI Assistants

1. AI assistants are computer programs designed to respond to user requests in a conversational manner.
2. AI assistants can be used for a variety of tasks, such as providing information, scheduling appointments, making recommendations, and more.
3. AI assistants can be programmed to be respectful, polite, and inclusive, and to follow certain guidelines such as avoiding self-harm, adult content, illegal activities, politics, specific financial advice, product recommendations, and celebrities.
4. AI assistants can be programmed to use natural language processing (NLP) to understand user requests and provide appropriate responses.
5. AI assistants can be programmed to use machine learning (ML) to improve their responses over time, by learning from user interactions.
6. AI assistants can be integrated into existing systems, such as customer service platforms, to provide a more automated and personalized experience.

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1. It is important to be aware of the potential risks associated with any kind of online activity.
2. Always use caution when providing personal information online, such as your full name, address, or credit card number.
3. Be mindful of the potential for cyberbullying, especially when using social media platforms.
4. Be aware of the potential for online predators and take steps to protect yourself.
5. Be aware of potential scams, such as phishing emails or online offers that seem too good to be true.
6. Be aware of the potential for data breaches or identity theft.
7. Be aware of the potential for online fraud and take steps to protect yourself.
8. Be aware of the potential for malicious software, such as viruses or spyware, and take steps to protect yourself.
9. Be aware of the potential for cyberstalking and take steps to protect yourself.
10. Be mindful of the potential for online harassment and take steps to protect yourself.

Topic: The Advantages of Artificial Intelligence

1. Improved Accuracy: AI systems can perform tasks more accurately than humans. This is especially beneficial in fields such as healthcare, where accuracy is critical.

2. Increased Efficiency: AI systems can complete tasks faster than humans, allowing businesses to become more efficient.
3. Reduced Costs: AI can reduce costs by automating tasks that would otherwise require manual labor.
4. Improved Decision Making: AI systems can analyze large amounts of data more quickly and accurately than humans, allowing businesses to make better decisions.
5. Increased Productivity: AI systems can automate tasks, freeing up humans to focus on more complex tasks. This can lead to increased productivity and output.
6. Improved Customer Service: AI can be used to provide customers with a more personalized experience, leading to increased customer satisfaction.

ENGINEERING PHYSICS LAB KCS

1. Engineering Physics is an interdisciplinary field which combines the principles of physics and engineering.
2. It is used to solve complex problems in areas such as energy, materials science, electronics, and biotechnology.
3. In the KCS Engineering Physics Lab, students are provided with a variety of tools and techniques to help them understand the fundamentals of engineering physics.
4. The lab provides students with access to a wide range of equipment such as oscilloscopes, signal generators, and computers.
5. Students can also use the lab to design and build their own electronic circuits and systems.
6. The lab also provides students with the opportunity to explore the principles of engineering physics through hands-on experiments.
7. In addition to the lab, students can also access online resources such as lecture notes, tutorials, and simulations.
8. These resources help students to gain a deeper understanding of engineering physics and its applications.

Course Objectives:

1. Understand the principles and fundamentals of programming
2. Develop an understanding of the core concepts of programming languages
3. Learn the fundamentals of data structures and algorithms
4. Develop basic problem-solving and debugging skills
5. Develop an understanding of the principles of software engineering
6. Develop an understanding of the principles of computer networks
7. Learn the fundamentals of web development

8. Develop an understanding of the principles of database design and management
9. Develop an understanding of the principles of user interface design
10. Develop an understanding of the principles of operating systems and computer architecture

1. To enable the students to understand about the fundamental concepts of analytical instruments

- Analytical instruments are used to measure and analyze physical and chemical properties of substances.
- Analytical instruments can be used to measure properties such as temperature, light, pressure, and chemical composition.
- Analytical instruments are used in a variety of fields such as biochemistry, geology, and engineering.
- Common analytical instruments include spectrophotometers, chromatographs, and mass spectrometers.
- Analytical instruments are used to analyze samples of materials to determine their composition and properties.
- Analytical instruments are used to detect and measure the presence of various substances in a sample.
- Analytical instruments can be used to measure physical properties such as temperature, pressure, and light.
- Analytical instruments are used to measure chemical properties such as pH, concentrations, and molecular structure.
- Analytical instruments are used to analyze the structure and composition of a sample.
- Analytical instruments are used to identify and quantify unknown substances in a sample.

To enable the students to understand about the analysis of chloride content, hardness, alkalinity

1. Chloride content can be determined by titration with silver nitrate solution. The end point of the titration is determined by the presence of a silver chloride precipitate.
2. Hardness of water can be determined by titration with EDTA. The end point of the titration is indicated by the presence of a pink-violet color.
3. Alkalinity of water can be determined by titration with hydrochloric acid. The end point of the titration is indicated by the presence of a yellow color.

Water

1. Water is a transparent, tasteless, odorless, and nearly colorless chemical substance, which is the main constituent of Earth's hydrosphere and the fluids of most living organisms.
2. It is vital for all known forms of life, even though it provides no calories or organic nutrients. Its chemical formula is H_2O , meaning that each of its molecules contains one oxygen and two hydrogen atoms, connected by covalent bonds.
3. Water is the name of the liquid state of H_2O at standard ambient temperature and pressure. It forms precipitation in the form of rain and aerosols in the form of fog.
4. Clouds are formed from suspended droplets of water and ice, its solid state. When finely divided, crystalline ice may precipitate in the form of snow.
5. The gaseous state of water is steam or water vapor. Water moves continually through the water cycle of evaporation, transpiration (evapotranspiration), condensation, precipitation, and runoff, usually reaching the sea.
6. Water covers 71% of the Earth's surface, mostly in seas and oceans. Small portions of water occur as groundwater, in the glaciers and the ice caps of Antarctica and Greenland, and in the air as vapor, clouds (formed of ice and liquid water suspended in air), and precipitation.
7. Water plays an important role in the world economy. Approximately 70% of the freshwater used by humans goes to agriculture.

To enable the students to understand about the measure of pH, surface tension and viscosity of

- pH is a measure of the acidity or alkalinity of a solution, usually measured on a scale from 0 to 14. A pH of 7 is considered neutral, a pH below 7 is acidic, and a pH above 7 is alkaline.
- Surface tension is the measure of the cohesive forces between molecules of a liquid that cause it to act as if its surface were a stretched elastic membrane.
- Viscosity is a measure of a fluid's resistance to flow. It is determined by the amount of internal friction between the molecules of a fluid and is affected by temperature, pressure and the type of fluid.

A Liquid

A liquid is a state of matter characterized by a lack of fixed shape and an ability to flow. It is one of the four fundamental states of matter, along with solids, gases, and plasma. A liquid is a nearly incompressible fluid that conforms to the shape of its container but retains a (nearly) constant volume independent of pressure.

Some of the most common liquids include water, oil, alcohol, and juice. Liquids are composed of tiny vibrating particles of matter, such as atoms, held together by intermolecular bonds. Like solids, the molecules in a liquid have a regular structure and move at different speeds.

Properties of a Liquid

- Liquids can flow and take the shape of their container.
- The volume of a liquid remains constant, regardless of pressure.
- The surface tension of a liquid is the result of the cohesive forces between molecules.
- The viscosity of a liquid is the measure of its resistance to flow.
- The boiling point of a liquid is the temperature at which it vaporizes.
- The freezing point of a liquid is the temperature at which it solidifies.

4. To enable the students to understand about the preparation of different resins

1. Resins are a type of natural or synthetic polymer made up of many repeating units linked together.
2. Natural resins are usually produced by plants, while synthetic resins are produced in a laboratory.
3. Resins can be used for a variety of applications, including coatings, adhesives, paints, and plastics.
4. Preparation of different resins involves the selection of the appropriate monomers, polymerization of the monomers, and post-polymerization treatments.
5. The selection of the monomers is based on the desired properties of the final product.
6. The polymerization process may involve either a chemical or physical reaction, depending on the type of monomers used.
7. Post-polymerization treatments may include curing, cross-linking, and other treatments to improve the properties of the resin.

Synthesis of Organic Compounds Such as Adipic

1. Organic compounds are molecules composed of carbon and hydrogen atoms, with other atoms such as oxygen, nitrogen, and sulfur often present.
2. Organic compounds can be divided into two main classes: aliphatic compounds, which contain only carbon and hydrogen atoms, and aromatic compounds, which contain carbon and hydrogen atoms arranged in a ring structure.
3. Adipic acid is an aliphatic dicarboxylic acid, which is an important industrial chemical used in the manufacture of nylon and other polymers.

4. Adipic acid is synthesized by the oxidation of cyclohexanol, which is obtained from the oxidation of cyclohexane.
5. The oxidation of cyclohexanol is carried out in the presence of a catalyst, such as manganese dioxide, at a temperature of around 160°C.
6. The oxidation of cyclohexanol results in the formation of cyclohexanone, which is then converted to adipic acid by a series of reactions known as the Kolbe-Schmitt reaction.
7. The Kolbe-Schmitt reaction involves the conversion of cyclohexanone to cyclohexanecarboxylic acid, which is then further oxidized to adipic acid.
8. Adipic acid can also be synthesized by the oxidation of cyclohexene, which is obtained from the dehydration of cyclohexanol.
9. The oxidation of cyclohexene is carried out in the presence of a catalyst, such as manganese dioxide, at a temperature of around 160°C.
10. The oxidation of cyclohexene results in the formation of cyclohexane-1,4-dione, which is then converted to adipic acid by a series of reactions known as the Beckmann rearrangement.
11. The Beckmann rearrangement involves the conversion of cyclohexane-1,4-dione to cyclohexane-1,2-dicarboxylic acid, which is then further oxidized to adipic acid.

Acid and Paracetamol by Conventional and Green Route

- Acid and paracetamol can be synthesized by conventional and green routes.
- The conventional route involves the use of a strong acid, such as sulfuric acid, to convert paracetamol into its acid form.
- The green route utilizes an enzyme-catalyzed reaction to convert paracetamol into its acid form.
- The green route is more environmentally friendly and uses fewer resources than the conventional route.
- The green route also produces fewer byproducts, which can be more easily disposed of.
- The green route is more expensive than the conventional route, but is becoming more cost-effective as technology advances.

List of Experiments

1. The Mpemba Effect: This experiment studies the phenomenon in which hot water can freeze faster than cold water.
2. The Boyle's Law Experiment: This experiment studies the relationship between the pressure and volume of a gas.
3. The Bernoulli's Principle Experiment: This experiment studies the rela-

tionship between the pressure, velocity, and height of a fluid.

4. The Archimedes' Principle Experiment: This experiment studies the relationship between the buoyant force and the volume of an object submerged in a fluid.
5. The Faraday's Law of Induction Experiment: This experiment studies the relationship between the electric current and the magnetic field.
6. The Ohm's Law Experiment: This experiment studies the relationship between the current and voltage in an electrical circuit.
7. The Doppler Effect Experiment: This experiment studies the phenomenon in which the frequency of a sound wave changes as the source moves relative to the observer.

1. Calibration of Analytical Equipment and Apparatus

1. Calibration is the process of ensuring that analytical equipment and apparatus are functioning correctly and accurately.
2. It is important to regularly check the accuracy of the equipment and apparatus, as they may become inaccurate over time due to wear and tear, changes in temperature, humidity, or other environmental factors.
3. Calibration should be performed in accordance with standard operating procedures (SOPs) and should be documented.
4. The calibration process typically involves comparing the instrument output to a known reference standard.
5. If the instrument is found to be inaccurate, adjustments may be made to bring it back into accuracy.
6. Calibration should be performed by qualified personnel and should be documented in the instrument logbook.
7. It is important to regularly check the accuracy of the equipment and apparatus, as they may become inaccurate over time due to wear and tear, changes in temperature, humidity, or other environmental factors.
8. Calibration should be performed in accordance with standard operating procedures (SOPs) and should be documented.

2. Determination of Hardness of Water Sample by EDTA Method

1. Hardness of water is an important physical property that is determined by the concentration of Ca^{2+} and Mg^{2+} ions present in water.
2. EDTA (Ethylenediaminetetraacetic acid) is a complexing agent used to determine the hardness of water by titration.

3. The EDTA titration method involves the addition of EDTA in a known concentration to a sample of water containing Ca^{2+} and Mg^{2+} ions, which then react with the EDTA to form a complex.
4. The end-point of the titration is determined by the use of an indicator, such as Eriochrome Black T, which changes color when all of the Ca^{2+} and Mg^{2+} ions have been complexed.
5. The total hardness of the water sample can then be calculated by multiplying the concentration of EDTA used in the titration by the volume of the sample.

3. Determination of Alkalinity of Water Sample

- Alkalinity is the measure of the capacity of water to resist changes in pH that would make the water more acidic.
- Alkalinity is measured in milligrams per liter (mg/L) of calcium carbonate (CaCO_3).
- Alkalinity is an important parameter for water quality and can be used to assess the water's ability to resist changes in pH.
- Alkalinity can be determined by titration with a strong acid, such as hydrochloric acid (HCl).
- The titration is performed by adding small amounts of acid to a sample of water until the pH of the water drops below a certain point.
- The amount of acid added is proportional to the alkalinity of the water sample.
- The alkalinity of the sample can be determined by measuring the amount of acid added and then calculating the alkalinity from the results.

4. Determination of pH by titrimetric method

1. Titrimetric methods are used to determine the pH of a solution.
2. This method involves titrating a solution of known concentration with a standard solution of known concentration until the endpoint of the reaction is reached.
3. The pH of the solution is then determined by the volume of the standard solution used in the titration.
4. The endpoint of the titration is determined by using an indicator, which changes color when the reaction is complete.
5. The endpoint can also be determined by using a pH meter, which measures the pH of the solution during the titration.
6. This method is accurate and reliable and can be used to determine the pH of a variety of solutions.

5. Determination of Surface Tension of a Given Liquid

1. Surface tension is a property of liquids that causes them to resist an external force, acting to draw the liquid's surface into a concave shape.

2. Surface tension can be measured by a variety of methods, including the use of a surface tensiometer, a device that measures the force required to pull a needle through a given liquid.
3. To measure the surface tension of a given liquid, the needle of the surface tensiometer is pulled through the liquid at a constant speed.
4. The force required to pull the needle through the liquid is then measured and used to calculate the surface tension of the liquid.
5. The surface tension of a given liquid can also be determined by measuring the maximum height of a liquid column that can be supported by the liquid's surface tension. This is known as the capillary rise method.
6. Knowing the surface tension of a given liquid can be useful for a variety of applications, such as the design of medical devices and the study of the behavior of liquids in various environments.

6. Determination of Viscosity of a given liquid by viscometer

1. Viscometers are instruments used to measure the viscosity of a liquid.
2. The viscosity of a liquid is a measure of its resistance to flow.
3. The viscosity of a liquid can be determined by measuring the time it takes for a given volume of liquid to flow through a capillary tube.
4. The viscosity of a given liquid can also be determined by using a viscometer.
5. A viscometer consists of a cup, a spindle and an adjustable weight.
6. The cup is filled with the liquid to be tested and the spindle is placed in the cup.
7. The adjustable weight is then placed on the spindle, and the time it takes for the spindle to reach a certain depth is measured.
8. The viscosity of the liquid is then calculated based on the time it took for the spindle to reach the desired depth.

7. Determination of the Strength of Ferrous Ammonium Sulfate using External Indicator

1. Ferrous ammonium sulfate is a type of compound used in laboratory experiments and other scientific applications.
2. It is important to accurately determine the strength of ferrous ammonium sulfate, as this can affect the outcome of an experiment.
3. An external indicator can be used to determine the strength of ferrous ammonium sulfate. This involves adding a known amount of the indicator to a known amount of the ferrous ammonium sulfate solution and observing the color change.

4. The color change will indicate the concentration of the ferrous ammonium sulfate solution. The color change will be more intense for higher concentrations.
5. The concentration of the ferrous ammonium sulfate solution can then be determined using a calibration curve.
6. It is important to note that the external indicator used must be compatible with the ferrous ammonium sulfate solution, as some indicators may not be suitable.

8. Determination of the Strength of Potassium Dichromate Using Internal Indicator

1. Potassium dichromate is a strong oxidizing agent and is commonly used in the titration of a reducing agent.
2. The strength of potassium dichromate can be determined by titrating it with a standard solution of a reducing agent, such as sodium thiosulfate.
3. The endpoint of the titration is usually determined using an external indicator, such as phenolphthalein.
4. An internal indicator can also be used to determine the endpoint of the titration.
5. The internal indicator is usually a solution of potassium iodide, which is added to the titration mixture at the start of the titration.
6. When the titration is complete, the solution will be orange in color, indicating that all of the potassium dichromate has been consumed.

9. Determination of Available Chlorine in Bleaching Powder

1. Bleaching powder is a white powder composed of calcium chloride, calcium hydroxide, and sodium hypochlorite.
2. It is used in the bleaching of textiles and paper products and as a disinfectant.
3. The available chlorine content of bleaching powder can be determined by titration with a standard solution of sodium thiosulfate.
4. A known volume of the bleaching powder solution is taken in a conical flask and titrated against the standard solution of sodium thiosulfate until the solution turns pale yellow.

5. The amount of sodium thiosulfate required for the titration is a measure of the available chlorine content of the bleaching powder.
6. The available chlorine content of the bleaching powder is calculated by multiplying the amount of sodium thiosulfate used for the titration with a factor of 0.086.

10. Determination of Chloride Content in Water Sample

1. Chloride is an anion composed of one chlorine atom and one anion. It is an essential component of the human body and is found in many natural waters.
2. The most common method for determining the chloride content in a water sample is the titration method. This method requires a titrant, which is a solution of known concentration, and an indicator, which changes color when the endpoint of the titration is reached.
3. The titration method is performed by adding the titrant to the water sample until the indicator changes color. The amount of titrant added is then used to calculate the concentration of chloride ions in the water sample.
4. Another method for determining the chloride content in a water sample is the gravimetric method. This method involves the precipitation of chloride ions with silver nitrate, followed by the filtration and drying of the precipitate. The mass of the precipitate is then used to calculate the chloride concentration in the water sample.
5. The spectrophotometric method is also used to determine the chloride content in a water sample. This method involves the addition of a reagent to the water sample, which reacts with the chloride ions to produce a colored complex. The concentration of the chloride ions is then determined by measuring the intensity of the color produced.
6. The chloride content in a water sample can also be determined by ion chromatography. This method involves the separation of the ions present in the water sample, followed by the determination of the concentration of chloride ions.

11. Preparation of Phenol formaldehyde (PF) Resin

1. Phenol formaldehyde (PF) resin is a thermosetting plastic material produced by the reaction of phenol and formaldehyde in the presence of an

acid catalyst.

2. The reaction is exothermic and the product is a viscous liquid.
3. The reaction is usually carried out in aqueous solution and the pH is maintained at around 4.5 by the addition of acid.
4. The reaction is complete when the viscosity of the reaction mixture reaches a predetermined value.
5. The product is then neutralized with a base and the pH is adjusted to around 7.
6. The product is then cooled and moulded into the desired shape.
7. The product can then be cured at an elevated temperature to give the desired properties.

12. Preparation of Urea Formaldehyde (UF) Resin

1. Urea formaldehyde (UF) resin is a synthetic polymer produced from urea and formaldehyde.
2. It is a thermosetting plastic, meaning it does not melt when heated but instead decomposes.
3. UF resin is used in many industries, such as adhesives, coatings, and laminates.
4. The preparation of UF resin involves combining urea and formaldehyde in an alkaline medium.
5. The reaction of these two components produces a polymer that is insoluble in water and has a high molecular weight.
6. The polymerization process is usually carried out at temperatures between 80°C and 110°C.
7. The reaction can also be accelerated by adding catalysts such as ammonia, sodium hydroxide, or sulfuric acid.
8. The UF resin can be cured further by heating it to temperatures between 150°C and 180°C.
9. The final product is a hard, rigid material that is resistant to water, acids, and alkalis.

13. Preparation of Adipic acid / Paracetamol

1. Adipic acid is an organic compound used in the production of polymers, such as nylon and polyurethane. It can also be used in the production of pharmaceuticals, food additives, and other industrial products.

2. Paracetamol is an analgesic drug used to treat mild to moderate pain and fever. It is available in various forms, including tablets, capsules, and liquids.
3. Adipic acid and paracetamol can be prepared through a combination of chemical reactions. The first step involves the reaction of an aldehyde with ammonia to form an imine. This imine is then reacted with an acid chloride to form an amide.
4. The amide is then reacted with acetic anhydride to form a carboxylic acid. This acid is then reacted with hydrazine to form adipic acid.
5. To prepare paracetamol, the adipic acid is then reacted with phenol in the presence of an acid catalyst. This reaction produces paracetamol and carbon dioxide.

14. Determination of Cell Conductance of a Solution

1. Cell conductance is a measure of the ability of a solution to conduct electrical current.
2. It is the reciprocal of the resistance of the solution, which is the opposition to the flow of electric current.
3. Cell conductance is usually measured in siemens (S) or millisiemens (mS).
4. Cell conductance can be determined by measuring the voltage across a known resistance in the solution.
5. The cell conductance of a solution is affected by factors such as temperature, concentration, and the presence of ions in the solution.
6. The cell conductance of a solution can be used to calculate the equivalent conductance of the ions in the solution.
7. The equivalent conductance of the ions in a solution is the sum of the individual ionic conductances of each ion present in the solution.
8. The total cell conductance of a solution is the sum of the individual ionic conductances, plus the contribution of the solvent.

15. Determination of Rate constant of hydrolysis of esters

1. Hydrolysis is a reaction where an ester is broken down into an alcohol and an acid by the addition of water.
2. The rate constant of hydrolysis is the rate at which the reaction occurs and is dependent on the type of ester and the reaction conditions.
3. The rate constant can be determined by measuring the rate of disappearance of the ester over time.
4. The rate of disappearance is proportional to the concentration of the ester and the rate constant.

5. The rate constant can be determined by measuring the rate of disappearance of the ester over time at different temperatures and then plotting the data on a graph.
6. The rate constant can also be determined by measuring the rate of disappearance of the ester over time at different concentrations of the ester and then plotting the data on a graph.
7. The rate constant can also be determined by measuring the rate of disappearance of the ester over time at different pH levels and then plotting the data on a graph.
8. The rate constant can also be determined by measuring the rate of disappearance of the ester over time at different concentrations of the acid and then plotting the data on a graph.
9. The rate constant can also be determined by measuring the rate of disappearance of the ester over time at different concentrations of the alcohol and then plotting the data on a graph.
10. The rate constant can also be determined by measuring the rate of disappearance of the ester over time at different concentrations of the water and then plotting the data on a graph.

16. Element Detection and Identification of Functional Groups in Organic Compounds

- Element detection in organic compounds is a process used to identify the elements present in a compound. This is usually done by analyzing the compound's molecular structure or its chemical properties.
- Functional groups are specific arrangements of atoms that are responsible for the majority of a compound's chemical behavior. These groups are typically composed of atoms from the same element, such as carbon, oxygen, nitrogen, and hydrogen.
- Element detection and identification of functional groups in organic compounds is typically done by spectroscopy, which is a technique that uses light or other forms of radiation to analyze the chemical structure of a compound.
- Spectroscopy can be used to identify the elements present in a compound and to determine the functional groups present. This information can be used to determine the compound's structure, reactivity, and other properties.
- The most common types of spectroscopy used in element detection and identification of functional groups in organic compounds are infrared spectroscopy, nuclear magnetic resonance spectroscopy, and mass spectrometry.
- By using spectroscopy, chemists can quickly and accurately determine the elements present in a compound and the functional groups that are responsible for its chemical behavior. This information can be used to better

understand the properties of a compound and to design new compounds with specific properties.

Experiments

1. Conduct an experiment to measure the speed of sound in air.
2. Measure the acceleration due to gravity experimentally.
3. Investigate the relationship between current and potential difference in a resistor.
4. Determine the resistance of a given wire by plotting a graph of potential difference versus current.
5. Determine the wavelength of light emitted by a sodium lamp.
6. Investigate the relationship between the length of a wire and its resistance.
7. Determine the frequency of an AC supply.
8. Investigate the effect of temperature on the resistance of a given material.
9. Measure the capacitance of a capacitor.
10. Investigate the effect of frequency on the power factor of an AC circuit.

Course Outcomes:

- Understand the fundamentals of computer systems and their components
- Analyze the architecture and design of computer systems
- Design and implement computer systems and their components
- Evaluate computer systems for performance and security
- Develop computer systems for a variety of applications
- Utilize computer systems in a professional setting
- Understand the principles of data science and machine learning
- Analyze and interpret data using statistical methods
- Create and implement data models for predictive analytics
- Utilize data visualization techniques to communicate insights
- Design and develop machine learning algorithms
- Develop and deploy machine learning applications
- Assess the ethical implications of data-driven decision making

Course Outcomes Bloom's

- Understand the fundamentals of Bloom's Taxonomy and the different levels of cognitive thinking

- Identify the different cognitive processes and skills that can be applied to different types of tasks
- Analyze the different levels of Bloom's Taxonomy and the associated cognitive processes
- Evaluate the effectiveness of different levels of Bloom's Taxonomy and the associated cognitive processes
- Demonstrate the ability to apply Bloom's Taxonomy and the associated cognitive processes to different types of tasks
- Create and implement a plan that effectively applies Bloom's Taxonomy and the associated cognitive processes to different types of tasks

Level

1. A level is a type of measurement used to determine the height or depth of an object, person, or other entity.
2. Levels can be used to measure the vertical distance between two points, or to measure the angle of a slope.
3. Levels are commonly used in construction, engineering, surveying, and other fields.
4. Levels typically consist of an elongated body with a bubble vial attached to it.
5. The bubble vial is used to measure the angle of the level relative to a flat surface.
6. Levels are usually calibrated to indicate a level surface when the bubble is centered in the vial.
7. Levels can also be used to measure the relative height of objects with respect to one another.

CO-1 Get an understanding of the use of different analytical instruments. K3

1. Analytical instruments are used to measure, analyze, and interpret physical and chemical properties of substances.
2. Examples of analytical instruments include spectrophotometers, chromatographs, refractometers, and microscopes.
3. Spectrophotometers measure the amount of light that is absorbed or emitted by a sample.
4. Chromatographs are used to separate and identify components of a sample.
5. Refractometers measure the refractive index of a sample, which can be used to identify and quantify substances.
6. Microscopes are used to observe and identify particles in a sample.
7. Analytical instruments are used in a variety of fields, including medicine, chemistry, and engineering.

CO : Measuring Molecular and System Properties such as Surface Tension

1. CO is a molecular compound composed of two oxygen atoms and one carbon atom.
2. The molecular weight of CO is 44.01 g/mol.
3. The boiling point of CO is -78.5 °C.
4. The surface tension of CO can be measured using a tensiometer.
5. The surface tension of CO is approximately 24.2 mN/m at room temperature.
6. The surface tension of CO is affected by temperature and pressure, so these variables must be taken into account when measuring surface tension.
7. The surface tension of CO can also be measured using a Wilhelmy plate method or a pendant drop method.
8. The surface tension of CO can also be measured using a bubble pressure method.
9. The surface tension of CO can be used to calculate the surface area of a liquid.

Viscosity, Conductance of Solution, Chloride and Iron Content in Water

- Viscosity is a measure of a fluid's resistance to flow. It is defined as the ratio of shear stress to the rate of shear strain. Water has a relatively low viscosity, meaning it flows easily.
- Conductance is a measure of the ease with which electricity can flow through a solution. It is the reciprocal of resistance and is usually measured in siemens per meter (S/m). The conductance of a solution is affected by the presence of ions, such as chloride and iron, which can increase electrical conductivity.
- Chloride is an anion found in many natural waters, and its concentration can affect the conductivity of a solution. Chloride concentrations in water can range from 0.1 to 500 mg/L.
- Iron is another ion found in natural waters. It is present in both ferrous (Fe²⁺) and ferric (Fe³⁺) forms. Iron can affect the taste and color of water, as well as its electrical conductivity. Iron concentrations in water can range from 0.01 to 20 mg/L.

K3

1. K3 stands for Knowledge, Skills, and Attitudes. It is a way of understanding and assessing the knowledge, skills, and attitudes that people need to be successful in their roles.

2. Knowledge is the information that people need to understand the context in which they are working and to make decisions.
3. Skills are the abilities that people need to be able to do their work. These skills can include technical skills, interpersonal skills, problem-solving skills, and communication skills.
4. Attitudes are the beliefs and values that people have about their work and the environment in which they are working. Attitudes can affect how people approach their work and how they interact with others.
5. K3 is an important concept for understanding and assessing the knowledge, skills, and attitudes that people need to be successful in their roles. It can be used to identify gaps in knowledge, skills, and attitudes and to develop strategies for addressing those gaps.

CO-3 Measure the hardness and alkalinity of the water. K3

1. Hardness of water is the measure of the concentration of calcium and magnesium ions in the water.
2. Alkalinity of water is the measure of the concentration of bicarbonate, carbonate, hydroxide and other alkaline substances in the water.
3. Hardness and alkalinity of water can be measured with a titration test using a titration kit.
4. The titration process involves adding drops of a reagent to the water sample until a color change is observed.
5. The number of drops of reagent added to the water sample is then used to calculate the hardness and alkalinity of the water.
6. Hardness and alkalinity of water can also be measured with a water hardness test strip.
7. The test strip will change color when dipped into the water sample, and the color change can be compared to a chart to determine the hardness and alkalinity of the water.

CO-4 Know the fundamental concepts of the preparation of phenol

1. Phenol is a compound consisting of a hydroxyl group (-OH) attached to an aromatic hydrocarbon group.
2. It is a colorless, water-soluble solid with a characteristic odor.
3. It is produced commercially by the oxidation of benzene with air.
4. The oxidation of benzene is carried out in the presence of a catalyst, such as manganese dioxide, and an oxidizing agent, such as nitric acid.
5. The reaction produces a mixture of phenol and other oxidation products, such as cresols and nitrophenols.
6. Phenol can also be prepared by the direct oxidation of cumene with air.
7. This process is known as the cumene process, and it is the most widely used method for the industrial production of phenol.

8. Phenol can also be prepared by the hydrolysis of chlorobenzene in the presence of an acid catalyst.
9. This process is known as the Kolbe-Schmitt reaction.
10. Phenol can also be prepared by the hydrolysis of phenylmagnesium bromide or by the oxidation of cyclohexanol.

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

- Formaldehyde is an organic compound composed of a single carbon atom surrounded by two hydrogen atoms. It is a colorless, flammable gas with a pungent odor. It is used in the production of plastics, resins, and other materials.
- Urea formaldehyde resin is a thermoset plastic made from urea, formaldehyde, and other chemicals. It is used in a wide variety of applications including furniture, insulation, and adhesives.
- Adipic acid is a white, crystalline organic acid with a sour taste. It is used in the production of nylon, polyurethanes, and other polymers.
- Paracetamol is an analgesic drug used to treat pain and fever. It works by blocking the production of certain chemicals in the body that cause inflammation and pain.

K3

1. K3 is an abbreviation for the 3rd generation of the Kelvin scale, a temperature scale used in science and engineering.
2. The Kelvin scale is based on the concept of absolute zero, the temperature at which all thermal motion ceases.
3. The Kelvin scale is an absolute temperature scale, meaning that temperatures on the Kelvin scale are independent of any external reference point.
4. The Kelvin temperature scale starts from absolute zero, which is defined as 0 K.
5. One degree Kelvin (1 K) is equal to one degree Celsius (1 °C).
6. The boiling point of water is 373.15 K (100 °C).
7. The freezing point of water is 273.15 K (0 °C).
8. The triple point of water, the temperature and pressure at which solid, liquid, and vapor phases of water can exist in equilibrium, is 273.16 K (0.01 °C).

CO-5 Estimate the rate constant of reaction. K3

1. The rate constant of a reaction, K3, is a measure of the speed of a chemical reaction.

2. It is a measure of how quickly a reaction takes place and is related to the activation energy of the reaction.
3. The rate constant is usually expressed as a reciprocal of time, such as seconds, minutes, or hours.
4. The rate constant can be calculated by measuring the change in the concentration of a reactant or product over time.
5. The rate constant can also be calculated by measuring the rate of a reaction at different temperatures and using the Arrhenius equation.
6. The rate constant can be used to predict the rate of a reaction at different temperatures.
7. The rate constant can also be used to compare the rates of different reactions.

Topic:

- 1.
- 2.
- 3.

Please provide a topic for me to write about.

Topic: The Benefits of Exercise

1. Exercise improves physical health by increasing heart and lung capacity, strengthening muscles, and reducing the risk of developing chronic diseases such as heart disease, stroke, and diabetes.
2. Exercise can also reduce stress, improve mood, and help to manage weight.
3. Regular physical activity can help to improve sleep quality and reduce fatigue.
4. Exercise can also improve cognitive functioning, including memory, concentration, and decision-making.
5. Exercise can help to build strong bones and reduce the risk of osteoporosis.
6. Exercise can also help to reduce anxiety and depression and improve self-esteem.
7. Regular physical activity can help to reduce the risk of certain types of cancer.
8. Exercise can also help to reduce the risk of developing chronic illnesses such as hypertension, high cholesterol, and type 2 diabetes.

Topic: The History of the Internet

1. The Internet first emerged in the late 1960s and early 1970s, when computer scientists developed the protocols and hardware necessary to connect

computers together.

2. The first public demonstration of the Internet was in 1972, when a researcher at the University of Hawaii connected two computers together over a long-distance telephone line.
3. The development of the Internet continued throughout the 1970s and 1980s, with the development of the Transmission Control Protocol (TCP) in 1974 and the Internet Protocol (IP) in 1983.
4. In 1985, the National Science Foundation (NSF) established the first large-scale network, called the NSFNET. This network connected universities, research institutions, and other government agencies.
5. The NSFNET was replaced by the modern Internet in 1995, when the NSFNET was decommissioned and commercial service providers began offering Internet access.
6. Since then, the Internet has become an integral part of modern life, with billions of people around the world using it for communication, commerce, entertainment, and more.

Topic: AI Assistants

1. AI assistants are computer programs designed to provide assistance to humans in a variety of tasks.
2. AI assistants use natural language processing to understand user commands and requests.
3. AI assistants can be used to answer questions, provide information, and even complete tasks.
4. AI assistants are becoming increasingly popular due to their ability to provide personalized and accurate responses.
5. AI assistants can be found in a variety of applications, from virtual assistants to smart home devices.
6. AI assistants can be used to automate mundane tasks, such as scheduling meetings or making reservations.
7. AI assistants can also be used to provide customer service, such as providing product recommendations or answering questions.
8. AI assistants are becoming more advanced, with the ability to learn from user interactions and provide more personalized responses.

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1. This page will provide an overview of the key concepts related to AI assistants.
2. AI assistants are computer programs that are designed to provide assistance to humans in a variety of tasks.
3. AI assistants can respond to voice commands, understand natural language, and provide personalized recommendations.

4. AI assistants can be used to automate mundane tasks, such as scheduling appointments, organizing emails, and managing calendars.
5. AI assistants can also be used to provide personalized customer service, such as answering questions, providing product recommendations, and resolving customer complaints.
6. AI assistants are becoming increasingly popular and are being used in a variety of industries, such as healthcare, finance, and retail.
7. AI assistants are also being used to provide more complex services, such as providing legal advice, conducting financial transactions, and providing medical advice.
8. AI assistants are becoming increasingly sophisticated and are able to provide more accurate and personalized services.
9. As AI assistants become more sophisticated, they will be able to provide more complex services and will become more integrated into our daily lives.

Topic: The Benefits of AI Technology

1. AI technology can automate mundane tasks, freeing up humans to focus on more creative tasks.
2. AI can help to optimize processes, resulting in greater efficiency and cost savings.
3. AI can provide insights into data that would be impossible or difficult for humans to uncover.
4. AI can be used to develop more personalized products and services.
5. AI can help to reduce human errors and increase accuracy.
6. AI can be used to detect and respond to potential security threats.
7. AI can be used to improve customer service and provide more accurate responses.
8. AI can help to reduce bias in decision making.

Engineering Physics Lab

1. Engineering Physics Lab is an important part of a physics curriculum. It is designed to give students hands-on experience in designing, building, and testing physical systems.
2. The lab typically consists of a series of experiments that demonstrate fundamental principles of physics. This can include experiments involving electricity, magnetism, optics, thermodynamics, and mechanics.
3. In the lab, students are expected to use their knowledge of physics to design, build, and test physical systems. This can involve building circuits, testing the properties of materials, or constructing simple machines.
4. Students should also be prepared to record their results and make obser-

vations about the experiments. This can help them gain an understanding of the principles of physics and how they apply to the real world.

5. The Engineering Physics Lab is an important part of a physics curriculum and should be taken seriously by students. It is important to read the instructions carefully and follow directions in order to get the most out of the lab.

Course Objectives:

- Understand the fundamentals of computer programming, including variables, data types, control structures, and functions.
- Develop an understanding of the principles of object-oriented programming.
- Become familiar with the syntax and semantics of the Java programming language.
- Develop the skills necessary to design and implement programs in Java.
- Learn best practices for coding, debugging, and testing.
- Develop the ability to analyze and solve complex programming problems.
- Develop the skills necessary to communicate effectively about programming topics.

1. To enable the students to understand about the fundamental concepts of analytical instruments

1. Analytical instruments are tools used to measure physical properties of a sample, such as its composition, structure, and other characteristics.
2. These instruments are used in a wide range of fields, such as chemistry, biology, physics, and engineering.
3. Analytical instruments can be used to measure a variety of properties, including chemical composition, physical properties, and electrical properties.
4. Different analytical instruments are used to measure different properties. For example, chromatography is used to measure the chemical composition of a sample, while spectroscopy is used to measure the physical and electrical properties of a sample.
5. Analytical instruments are typically used in laboratories, where they are operated by trained professionals.
6. The accuracy and precision of an analytical instrument depend on the quality of the instrument and the skill of the operator.
7. It is important to understand the limitations of an analytical instrument and the conditions under which it should be used.

2. To enable the students to understand about the analysis of chloride content, hardness, alkalinity of water

1. **Chloride content:** Chloride is a common anion found in water. It is measured in milligrams per liter (mg/L) and is an indicator of water quality. High levels of chloride can be a sign of pollution or contamination.
2. **Hardness:** Hardness is a measure of the amount of calcium and magnesium dissolved in water. It is measured in milligrams per liter (mg/L) and is an indicator of water quality. High levels of hardness can cause scaling and staining of plumbing fixtures.
3. **Alkalinity:** Alkalinity is a measure of the buffering capacity of water. It is measured in milligrams per liter (mg/L) and is an indicator of water quality. High levels of alkalinity can cause corrosion of pipes and other plumbing fixtures.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of a liquid:

1. **pH** is a measure of the acidity or alkalinity of a solution. It is measured on a scale of 0 to 14, with 0 being the most acidic and 14 being the most alkaline.
2. **Surface tension** is a measure of the cohesion of molecules in a liquid. It is measured in dynes per centimeter (dyn/cm).
3. **Viscosity** is a measure of the internal friction of a liquid. It is measured in centipoise (cP).

4. To enable the students to understand about the preparation of different resins

- Resins are materials that are used to form strong and durable bonds between surfaces. They are often used in the construction and repair of furniture, automobiles, and other objects.
- Resins are typically made up of a combination of materials, including polymers, monomers, and other chemicals. The combination of these materials determines the properties of the resin and its ability to form a strong bond.
- Resins can be prepared in a variety of ways, depending on the desired properties of the final product. For example, some resins may be heated and cooled to form a solid material, while others may be mixed with solvents to form a liquid material.
- The preparation of different resins involves the selection of the appropriate materials, the correct combination of these materials, and the precise application of heat and/or pressure. It is important to understand the properties of the different materials and the desired outcome of the resin in order to achieve the desired results.

5. To enable the students to understand about the synthesis of organic compounds such as adipic acid and paracetamol by conventional and green route

- Synthesis of organic compounds such as adipic acid and paracetamol can be achieved through conventional and green route.
- Conventional synthesis of organic compounds involves the use of chemical reagents and solvents such as mineral acids and organic solvents. These processes generate hazardous by-products and can be energy intensive.
- Green synthesis, on the other hand, uses environmentally friendly reagents, catalysts and solvents, and is more cost-effective and energy-efficient.
- Adipic acid is an important industrial chemical used in the production of nylon and polyurethane. It can be synthesized through conventional route, which involves the oxidation of cyclohexanol with nitric acid, or through green synthesis, which uses aqueous hydrogen peroxide as oxidizing agent.
- Paracetamol is a widely used drug for the treatment of mild to moderate pain and fever. It can be synthesized through conventional route, which involves the reaction of phenol with acetic anhydride, or through green synthesis, which uses aqueous hydrogen peroxide as oxidizing agent.

LIST OF EXPERIMENTS

1. The Stanford Prison Experiment: This experiment was conducted in 1971 by psychologist Philip Zimbardo at Stanford University. It was designed to measure the psychological effects of becoming a prisoner or a guard.
2. The Milgram Experiment: This experiment was conducted in 1961 by psychologist Stanley Milgram at Yale University. It was designed to measure the willingness of participants to obey orders even when they conflict with their personal conscience.
3. The Asch Conformity Experiments: This experiment was conducted in 1951 by psychologist Solomon Asch at Swarthmore College. It was designed to measure the degree to which social pressure from a majority group could affect a person's opinion.
4. The Little Albert Experiment: This experiment was conducted in 1920 by psychologist John Watson at Johns Hopkins University. It was designed to measure the effects of classical conditioning on a child.
5. The Bystander Effect: This experiment was conducted in 1968 by psychologist John Darley and Bibb Latane at New York University. It was designed to measure the influence of the presence of others on a person's willingness to help in an emergency situation.

1. Calibration of Analytical Equipment and Apparatus

1. Calibration is the process of comparing a measurement instrument or system to a known standard to determine its accuracy and precision.

2. The purpose of calibration is to ensure that the instrument or system is providing accurate and reliable readings.
3. Analytical equipment and apparatus may include, but are not limited to, balances, pH meters, spectrophotometers, chromatographs, refractometers, and thermometers.
4. Calibration should be performed regularly in order to ensure accuracy and precision of the instrument or system.
5. The frequency of calibration depends on the type of instrument or system, the environment in which it is used, and the accuracy required.
6. Calibration should be performed by a qualified technician or engineer and should follow the manufacturer's instructions.
7. The calibration procedure should include a check of the instrument or system's accuracy, precision, linearity, and repeatability.
8. The results of the calibration should be documented and used to determine the performance of the instrument or system.
9. If the instrument or system fails to meet the requirements, it should be adjusted or repaired as necessary.

2. Determination of Hardness of Water Sample by EDTA Method

1. Hardness of water is a measure of the amount of dissolved calcium and magnesium ions present in the water.
2. EDTA (ethylenediaminetetraacetic acid) is a chelating agent that can be used to determine the hardness of a water sample.
3. EDTA forms a complex with the calcium and magnesium ions, preventing them from forming precipitates.
4. The amount of EDTA required to form a complex with the calcium and magnesium ions is proportional to the total hardness of the sample.
5. The EDTA method of determining water hardness involves titrating the water sample with a known concentration of EDTA solution.
6. The endpoint of the titration is determined by the use of an indicator, such as Eriochrome Black T, which changes color when the sample is free of calcium and magnesium ions.
7. The total hardness of the sample is then calculated by multiplying the volume of EDTA used in the titration by the concentration of the EDTA solution.

3. Determination of Alkalinity of Water Sample

1. Alkalinity is the ability of a water sample to neutralize an acid.
2. Alkalinity is measured in terms of milligrams per liter (mg/L) of calcium carbonate (CaCO_3).
3. Alkalinity can be determined by titration with a standard acid solution.

4. The endpoint of the titration is determined by a color indicator.
5. The total alkalinity of a water sample is calculated by subtracting the acid volume from the base volume.
6. The result is then multiplied by the equivalent weight of calcium carbonate to determine the alkalinity.

4. Determination of pH by Titrimetric Method

- Titrimetric methods involve the use of a titrant and an indicator to determine the end point of a titration.
- The titrant is a solution of known concentration, usually a base or an acid, which is added to a solution of unknown concentration.
- The indicator is a chemical that changes color when the solution reaches a certain pH.
- In titrimetric methods, the amount of titrant added is determined by measuring the pH of the solution at the end point of the titration.
- The end point of the titration is typically identified by the color change of the indicator.
- The most common type of titration used in the determination of pH is the acid-base titration.
- In acid-base titrations, the pH of the solution is determined by the amount of base that has been added to neutralize the acid.
- The end point of the titration is typically identified by the color change of the indicator.
- Another type of titration used in the determination of pH is the redox titration.
- In redox titrations, the amount of oxidizing or reducing agent added is determined by measuring the potential of the solution at the end point of the titration.
- The end point of the titration is typically identified by the color change of the indicator.
- The pH of the solution can also be determined by using a pH meter.
- The pH meter is a device that measures the electrical potential of a solution and converts it into a pH value.
- The pH meter is a more accurate method of determining pH than titrimetric methods.

5. Determination of surface tension of given liquid

1. Surface tension is a property of liquids that results from the cohesive forces between molecules.
2. Surface tension is responsible for the shape of liquid droplets, as well as for the ability of some insects (e.g. water striders) to walk on the surface of a body of water.
3. The surface tension of a given liquid can be determined using a variety of methods, including the use of a tensiometer, an instrument that measures

the force required to draw a liquid into a capillary tube.

4. The force required to draw the liquid into the tube is a measure of the surface tension.
5. Other methods of determining surface tension include the use of a ring, a drop weight, and a Wilhelmy plate.
6. In each of these methods, the force required to draw the liquid into a given shape is a measure of the surface tension.

6. Determination of Viscosity of a given liquid by viscometer

1. Viscosity is the measure of a fluid's resistance to flow.
2. A viscometer is an instrument used to measure the viscosity of a fluid.
3. The most common type of viscometer is the rotational viscometer, which measures the torque required to rotate a spindle immersed in the fluid.
4. To use a rotational viscometer, the spindle is immersed in the fluid and a known torque is applied to the spindle.
5. The spindle is then rotated at a constant speed and the torque required to maintain the rotation is measured.
6. The viscosity of the fluid is then calculated from the measured torque and the known parameters of the spindle.
7. The viscosity of a given liquid can also be determined by other methods, such as capillary viscometers, falling sphere viscometers, orifice viscometers, and rheometers.

7. Determination of the Strength of Ferrous Ammonium Sulfate Using External Indicator

1. Ferrous ammonium sulfate (FAS) is a chemical compound used in a variety of applications, including as an oxidizing agent, a reagent for organic synthesis, and a source of iron in nutrition.
2. FAS can be determined by titration with an external indicator. This method is based on a reaction between FAS and the indicator, which produces a color change that can be used to determine the concentration of FAS.
3. An appropriate indicator for FAS titration must be chosen. Commonly used indicators for FAS titration include methyl orange, phenolphthalein, and bromocresol green.
4. The endpoint of the titration is determined by the color change of the indicator. The endpoint is the point at which the indicator changes color and indicates the completion of the reaction.
5. The strength of the FAS solution can then be determined by calculating the molarity of the FAS solution. This is done by dividing the amount of FAS used in the titration by the volume of the solution.

8. Determination of the Strength of Potassium Dichromate Using Internal Indicator

1. Potassium dichromate is a strong oxidizing agent, and its strength can be determined by titration with a standard solution of a reducing agent, such as sodium thiosulfate.
2. The strength of potassium dichromate can also be determined using an internal indicator, such as methyl orange.
3. The internal indicator is added to the potassium dichromate solution, and the solution is titrated with the standard solution of sodium thiosulfate.
4. The end point of the titration is indicated by a change in the colour of the solution.
5. The strength of the potassium dichromate is calculated by using the formula: $\text{Strength} = (\text{Volume of sodium thiosulfate}) \times (\text{Molarity of sodium thiosulfate}) / (\text{Volume of potassium dichromate})$.

9. Determination of Available Chlorine in Bleaching Powder

- Bleaching powder is a white or yellowish powder composed of calcium hypochlorite ($\text{Ca}(\text{OCl})$).
- It is used as a disinfectant and bleaching agent in many industries.
- The available chlorine in bleaching powder can be determined by a titration method.
- The sample is dissolved in water and then titrated with a standard sodium thiosulphate solution.
- The end point of the titration is determined by the appearance of the yellow colour of iodine.
- The amount of available chlorine present in the sample is calculated from the volume of sodium thiosulphate used in the titration.
- The results obtained from this method are usually expressed as a percentage of available chlorine.

10. Determination of Chloride Content in Water Sample

1. Chloride can be determined in water samples using a variety of methods, including titration, potentiometric methods, and colorimetric methods.
2. Titration is a method that involves the addition of a known quantity of a reagent to a sample of water to determine the amount of chloride present.
3. Potentiometric methods involve the use of electrodes to measure the electrical potential of the sample, which can then be used to calculate the amount of chloride present.
4. Colorimetric methods involve the use of a colorimetric reagent that reacts with the chloride in the sample to produce a color change that can be measured.
5. In order to obtain accurate results, it is important to ensure that the sample is free of interfering substances and that the sample is prepared correctly.

6. The sample should be filtered to remove any solid particles that may interfere with the analysis.
7. The sample should also be acidified to ensure that any carbonates present are converted to chloride before analysis.
8. The sample should also be diluted to ensure that the concentration of chloride is within the range of the method being used.
9. Once the sample has been prepared, the method chosen should be followed to determine the amount of chloride present in the sample.
10. The results should then be recorded and the sample discarded in accordance with local regulations.

11. Preparation of Phenol formaldehyde (PF) Resin

1. Phenol formaldehyde (PF) resin is a thermosetting resin that is used in various applications such as adhesives, coatings, and laminates.
2. The resin is prepared by the reaction of phenol and formaldehyde in the presence of an acid or base catalyst.
3. The reaction is carried out in an aqueous solution at a temperature of 70-80°C.
4. The reaction is exothermic and the temperature must be controlled to prevent the reaction from getting out of control.
5. The reaction is complete when the mixture reaches a pH of 7.5.
6. The resin is then precipitated and washed with water to remove any unreacted reactants.
7. The resin is then dried and ground into a powder for use in various applications.

12. Preparation of Urea formaldehyde (UF) Resin

1. Urea formaldehyde (UF) resin is a thermosetting adhesive that is used for bonding wood and other materials.
2. It is made from urea and formaldehyde, which are reacted together in an acid solution.
3. The acid solution helps to catalyze the reaction between the two molecules, resulting in a cross-linked polymeric structure.
4. The urea and formaldehyde are mixed together in a ratio of 1:1.5 and heated to a temperature of 80-90°C.
5. The solution is then cooled and the resulting resin is ready for use.
6. UF resin is used in a variety of applications, including furniture, cabinetry, flooring, and wall paneling.
7. It is also used to make particleboard and medium-density fiberboard (MDF).
8. UF resin is known for its strength and durability, and is often used in place of other adhesives such as wood glue.

13. Preparation of Adipic acid / Paracetamol

1. Adipic acid is an organic compound that is used in the production of nylon. It is a white, crystalline solid with a slightly acidic taste.
2. Paracetamol is an analgesic and antipyretic drug used to treat mild to moderate pain and fever.
3. Adipic acid and paracetamol can be prepared from benzene, an aromatic hydrocarbon.
4. The first step in the preparation of adipic acid is the oxidation of benzene to benzoic acid. This is done by treating benzene with oxygen in the presence of a catalyst such as manganese dioxide.
5. The benzoic acid is then heated with concentrated nitric acid, forming adipic acid and water.
6. Paracetamol can be prepared by treating adipic acid with acetic anhydride and sodium acetate. This reaction produces paracetamol, acetic acid, and sodium adipate.
7. The products are then separated and purified. The paracetamol is then recrystallized to obtain a pure product.

14. Determination of Cell Conductance of a Solution

1. Cell conductance is a measure of the ability of an electrolyte solution to conduct electricity.
2. It is expressed as the reciprocal of the resistance of a solution to the flow of electricity, usually expressed in siemens per meter (S/m).
3. Cell conductance is determined by measuring the current passing through a solution of known volume and concentration.
4. The conductance of a solution is affected by the number of ions present in the solution, as well as the mobility of the ions.
5. The conductance of a solution can also be affected by temperature, pressure, and the concentration of other substances in the solution.
6. It is important to note that the cell conductance of a solution is not constant, and can vary with time.
7. The cell conductance of a solution can be used to measure the concentration of ions in a solution, as well as the rate of ion movement.
8. It can also be used to determine the rate of diffusion of ions in a solution.

15. Determination of Rate constant of Hydrolysis of Esters

1. Hydrolysis of esters is a chemical reaction in which an ester molecule is broken down into a carboxylic acid and an alcohol by the addition of a molecule of water.

2. The rate constant of hydrolysis of esters is a measure of the speed of the reaction and can be determined by measuring the change in concentration of the reacting species over time.
3. The rate constant of hydrolysis of esters can be determined experimentally by monitoring the disappearance of the ester and the appearance of the carboxylic acid and alcohol product.
4. The rate constant can also be determined using a mathematical model that takes into account the reactants, products, and rate constants of the reaction.
5. The rate constant of hydrolysis of esters can be used to predict the rate of reaction and the amount of product that will be produced.

16. Element detection and Identification of Functional Groups in Organic Compounds

1. **Element Detection:** Element detection involves the use of various chemical tests to identify the presence of specific elements in organic compounds. Common tests used for element detection include flame tests, acid-base reactions, and spectroscopy.
2. **Functional Group Identification:** Functional groups are specific groups of atoms that are responsible for the chemical properties of an organic compound. Common functional groups include alcohols, aldehydes, carboxylic acids, and amines. Functional group identification involves the use of various chemical tests to identify the presence of specific functional groups in organic compounds. Common tests used for functional group identification include color tests, infrared spectroscopy, and nuclear magnetic resonance spectroscopy.

Topic: Experiments

1. Conduct experiments to observe the effects of varying temperature on the rate of a chemical reaction.
2. Observe the effects of varying pressure on the rate of a chemical reaction.
3. Investigate the effects of varying concentrations of a reactant on the rate of a chemical reaction.
4. Study the effects of a catalyst on the rate of a chemical reaction.
5. Investigate the effects of varying pH on the rate of a chemical reaction.
6. Observe the effects of light intensity on the rate of a chemical reaction.
7. Analyze the effects of varying surface area on the rate of a chemical reaction.
8. Examine the effects of varying electric current on the rate of a chemical reaction.
9. Investigate the effects of varying magnetic field strength on the rate of a chemical reaction.
10. Analyze the effects of varying stirring speed on the rate of a chemical reaction.

Course Outcomes:

- Students will be able to explain the fundamentals of computer science and its major branches.
- Students will be able to understand the major components of a computer system and how they interact.
- Students will be able to analyze algorithms and evaluate their complexity.
- Students will be able to design and implement programs using appropriate programming languages.
- Students will be able to understand the principles of software engineering and apply them to the design and development of software.
- Students will be able to apply the principles of database design and management to create and manipulate databases.
- Students will be able to apply the principles of networking and distributed computing to the design and implementation of computer systems.
- Students will be able to understand the principles of artificial intelligence and apply them to the development of intelligent systems.

Upon completion of the course the student should be able to:

- Understand the fundamentals of computer programming
- Analyze and interpret data
- Design and develop algorithms
- Utilize problem-solving techniques
- Implement software solutions
- Test and debug programs
- Utilize best practices for software development

Course Outcomes Bloom's

1. Demonstrate an understanding of the fundamental concepts, theories, principles, and processes of a given subject or field.
2. Develop an understanding of the connections between the fundamental concepts, theories, principles, and processes of a given subject or field.
3. Demonstrate an ability to apply the fundamental concepts, theories, principles, and processes of a given subject or field.
4. Demonstrate an ability to analyze and evaluate the fundamental concepts, theories, principles, and processes of a given subject or field.
5. Demonstrate an ability to synthesize the fundamental concepts, theories, principles, and processes of a given subject or field.
6. Demonstrate an ability to create new knowledge based on the fundamental concepts, theories, principles, and processes of a given subject or field.

Level

1. Level is a term used to describe the height or depth of something. It can also be used to refer to a stage or degree of something.

2. In mathematics, a level is a line or surface that is perpendicular to a given direction. For example, a vertical level is a line that is perpendicular to the ground.
3. In physics, a level is a measurement of the energy of a system. For example, a particle at a higher level of energy will move faster than one at a lower level.
4. In engineering, a level is a tool used to measure the slope of a surface. It consists of a bubble that is suspended in a liquid-filled tube.
5. In architecture, a level is a device used to measure the height of a building. It consists of a plumb bob suspended from a string.
6. In business, a level is a measure of the success of a company or organization. It is typically measured by the amount of revenue or profit the company is generating.

CO-1 Get an Understanding of the Use of Different Analytical Instruments. K3

1. Analytical instruments are used to measure and analyze physical and chemical properties of different materials.
2. Examples of analytical instruments include spectrometers, chromatographs, and mass spectrometers.
3. Spectrometers measure the intensity of light emitted from a sample and can be used to identify elements and compounds in a sample.
4. Chromatographs separate and analyze the components of a sample, such as gas or liquid.
5. Mass spectrometers measure the mass of a sample and can be used to identify elements and compounds in a sample.
6. Analytical instruments are used in a variety of fields, including chemistry, biology, and medicine.

CO-2 Measure the Molecular/System Properties such as Surface Tension

1. Carbon dioxide (CO₂) is a colorless, odorless gas that is composed of one carbon atom and two oxygen atoms.
2. The molecular weight of CO₂ is 44.01 g/mol.
3. The boiling point of CO₂ is -78.5°C.
4. The melting point of CO₂ is -56.6°C.
5. The surface tension of CO₂ is 0.072 N/m.
6. The critical temperature of CO₂ is 31.1°C.
7. The critical pressure of CO₂ is 73.8 bar.
8. The critical density of CO₂ is 467.6 kg/m³.

9. The viscosity of CO₂ is 0.001 Pa · s.
10. The electrical conductivity of CO₂ is 0.0045 S/m.

Viscosity, Conductance of Solution, Chloride and Iron Content in the Water

- Viscosity is a measure of a fluid's resistance to flow. It is determined by the forces between molecules and is affected by temperature, pressure and the type of fluid.
- Conductance of a solution is a measure of how easily electricity flows through it. It is determined by the concentration of ions in the solution, as well as the type of ions present.
- Chloride is an anion found in many aqueous solutions. It is a major component of seawater and is commonly found in natural waters.
- Iron is an essential nutrient for many organisms and is found in many natural waters. It can be present in a variety of forms, including soluble and insoluble forms.

K3

1. K3 is a term used to refer to the third generation of the Internet. It is characterized by the emergence of new technologies such as the Internet of Things (IoT), Artificial Intelligence (AI), and blockchain.
2. K3 is expected to have a more secure, reliable, and efficient infrastructure than previous generations of the Internet. It will also be more decentralized, with data stored across multiple devices and locations.
3. K3 will enable new applications and services that are not possible with the current Internet, such as autonomous vehicles, smart homes, and health-care systems.
4. K3 will bring about changes in the way people interact with each other and with machines. It will also create opportunities for businesses to innovate and create new products and services.
5. K3 will require new security measures to protect data and ensure privacy. It will also require new regulations to ensure that the technology is used responsibly and ethically.

CO-3 Measure the hardness and alkalinity of the water. K3

1. Hardness of water is a measure of the amount of calcium and magnesium ions present in the water.
2. Alkalinity of water is a measure of the amount of carbonate, bicarbonate, hydroxide, and other alkaline substances present in the water.
3. Hardness and alkalinity of water can be measured using a titration method. This involves adding a standard acid solution to the water sample until the pH of the solution reaches a certain level.
4. The amount of acid required to reach the desired pH is then used to calculate the hardness and alkalinity of the water sample.

5. The hardness of water is typically expressed as milligrams per liter (mg/L) of calcium carbonate (CaCO_3).
6. Alkalinity of water is typically expressed as milligrams per liter (mg/L) of calcium carbonate (CaCO_3).
7. The hardness and alkalinity of water can affect the taste and odor of the water, as well as its suitability for different uses.

CO-4 Know the fundamental concepts of the preparation of phenol

1. Phenol is an aromatic organic compound, consisting of a hydroxyl group (-OH) attached to a benzene ring.
2. Phenol can be prepared by the oxidation of benzene, either by direct oxidation or by hydrolysis of benzene sulfonates.
3. The direct oxidation of benzene to phenol is usually done with air or oxygen in the presence of a catalyst such as manganese dioxide.
4. The hydrolysis of benzene sulfonates involves the reaction of benzene sulfonates with water in the presence of an acid catalyst.
5. Phenol can also be prepared from the hydrolysis of chlorobenzene in the presence of an acid catalyst.
6. Phenol can also be prepared from the dehydration of cresols in the presence of an acid catalyst.
7. Phenol can also be prepared from the oxidation of cumene in the presence of a catalyst such as manganese dioxide.
8. Phenol can also be prepared by the hydrolysis of phenol ethers.
9. Phenol can be purified by distillation and recrystallization.

Formaldehyde and Urea Formaldehyde Resin

- Formaldehyde is a colorless, flammable gas with a strong odor. It is used in the manufacture of plastics, resins, and other products. Urea formaldehyde resin is a thermosetting plastic made from urea and formaldehyde. It is used in the production of adhesives, coatings, and other products.

Adipic Acid

- Adipic acid is an organic compound with the formula $(\text{CH}_2)_4(\text{COOH})_2$. It is an odourless, white, crystalline solid that is soluble in water. Adipic acid is used in the production of nylon, polyurethane, and other products.

Paracetamol

- Paracetamol is an analgesic drug used to relieve pain and reduce fever. It is commonly used to treat headaches, muscle aches, and other minor aches and pains. Paracetamol is available in the form of tablets, capsules, and syrups.

K3

1. K3 is a type of artificial intelligence (AI) that is designed to emulate human behavior, such as understanding and responding to natural language, recognizing faces and objects, and making decisions.
2. K3 AI is used in a variety of applications, from customer service and robotics to healthcare and finance.
3. K3 AI systems are based on machine learning algorithms, which use data to build and improve models that can recognize patterns and make predictions.
4. K3 AI systems can be trained to recognize complex tasks, such as natural language processing (NLP), computer vision, and speech recognition, allowing them to understand and interact with humans in a more natural way.
5. K3 AI systems are used in a variety of industries, including healthcare, finance, retail, and manufacturing.
6. K3 AI systems are able to improve decision-making processes, automate mundane tasks, and provide more accurate insights into customer behavior.

CO-5 Estimate the rate Constant of Reaction. K3

1. The rate constant of a reaction is a measure of the rate at which reactants are converted to products.
2. It is usually expressed as the rate of the reaction per unit of concentration of the reactants.
3. The rate constant of a reaction is dependent on the temperature and the type of reaction.
4. For a given reaction, the rate constant can be calculated using the Arrhenius equation, which is given by:

$$K = Ae^{(-E_a/RT)}$$

Where:

A = Pre-exponential factor

E_a = Activation energy

R = Universal gas constant

T = Temperature in Kelvin

5. The rate constant can also be determined experimentally by measuring the rate of reaction at different temperatures.
6. The rate constant can then be used to calculate the rate of reaction at other temperatures.

7. Knowing the rate constant of a reaction is important in designing and controlling chemical processes, as well as predicting the products of a reaction.

Topic:

1. Here is the first point
2. Here is the second point
3. Here is the third point
4. Here is the fourth point
5. Here is the fifth point
6. Here is the sixth point

Topic:

- 1.
- 2.
- 3.
- 4.
- 5.

Please provide a more specific topic for me to address.

Topic: The Benefits of AI

AI (Artificial Intelligence) has become increasingly prevalent in our lives, from the way we shop to the way we communicate. Here are some of the benefits of AI:

1. Increased Efficiency: AI can complete complex tasks quickly and accurately, making it a valuable tool for businesses and organizations.
2. Improved Accuracy: AI can reduce the risk of human error by providing accurate and reliable data.
3. Reduced Costs: AI can reduce labor costs and improve efficiency, resulting in cost savings for businesses.
4. Enhanced Security: AI can detect potential threats and protect data from unauthorized access.
5. Improved Customer Service: AI can provide personalized customer service, allowing businesses to better meet customer needs.

6. Increased Productivity: AI can automate mundane tasks, freeing up time for employees to focus on more important tasks.
7. Improved Decision Making: AI can analyze large amounts of data quickly, helping businesses make more informed decisions.

The History of the Internet

1. The Internet began in the 1960s as a project of the United States Department of Defense, known as ARPANET.
2. In the 1970s, the idea of a global computer network was developed by a group of computer scientists, including Vint Cerf and Bob Kahn.
3. In the 1980s, the National Science Foundation (NSF) established the NSFNET, which allowed for the connection of many different networks.
4. In the 1990s, the World Wide Web (WWW) was invented by Tim Berners-Lee, and the first web browser, Mosaic, was released.
5. In the 2000s, broadband internet became widely available, and the number of users of the internet around the world began to skyrocket.
6. Today, the internet is used for a variety of purposes, including communication, entertainment, commerce, and education.

Topic:

This document provides an overview of the key concepts related to the topic.

1. Definition of the topic
2. History of the topic
3. Key elements of the topic
4. Examples of the topic
5. Impact of the topic
6. Future of the topic

Page 31 of 40

1. The topic of page 31 of 40 is the role of AI in healthcare.
2. AI technology can be used to improve patient care, reduce medical errors, and create more efficient and cost-effective healthcare systems.
3. AI can help doctors diagnose illnesses more accurately and quickly, as well as identify potential health risks before they become serious.
4. AI can also be used to automate mundane tasks such as filling out forms and scheduling appointments.

5. AI can be used to analyze large amounts of data to identify patterns and trends that can help predict and prevent diseases.
6. AI can also be used to create personalized treatments based on a patient's individual needs.
7. AI can help healthcare professionals make more informed decisions by providing them with more accurate and up-to-date information.
8. AI can also be used to create virtual assistants that can provide patients with advice and support.

Topic: The Advantages of AI Assistants

AI Assistants are becoming increasingly popular in our day-to-day lives. They are used in a variety of areas, from customer service to home automation. Here are some of the advantages of using AI Assistants:

- **Improved Efficiency:** AI Assistants are able to quickly and accurately process data and provide responses in a timely manner. This allows businesses to operate more efficiently and effectively.
- **Reduced Costs:** AI Assistants can reduce the costs associated with manual labor and customer service. By automating certain tasks, businesses can save money and time.
- **Improved Accuracy:** AI Assistants are able to provide more accurate responses than humans. This means that businesses can make more informed decisions and reduce errors.
- **Increased Security:** AI Assistants can help protect businesses from security threats by detecting and responding to malicious activity.
- **Increased Accessibility:** AI Assistants can make it easier for customers to access services and products. This can help businesses reach a wider audience and increase customer satisfaction.

ENGINEERING MATHEMATICS LAB KCS

1. Engineering mathematics is the application of mathematics to solve problems in engineering. It includes the use of mathematics to model and analyze engineering systems, as well as the development and application of mathematical techniques to solve engineering problems.
2. Engineering mathematics involves the use of mathematical models to solve engineering problems. This includes the use of mathematical techniques such as calculus, linear algebra, probability, and statistics.
3. The engineering mathematics lab is a laboratory designed to provide students with the opportunity to apply the mathematical techniques they have learned in a practical setting. The lab provides students with the op-

portunity to develop and apply mathematical models to solve engineering problems.

4. The lab includes a variety of equipment, such as computers, calculators, and software. Students use these tools to develop and analyze mathematical models and to solve engineering problems.
5. The lab also provides students with the opportunity to work in teams and collaborate on projects. This allows students to gain experience in working together to solve engineering problems.
6. The lab also provides students with the opportunity to interact with faculty and other students in the engineering mathematics program. This allows students to gain insight into the field of engineering mathematics and to learn from the experiences of others.

Course Objectives:

1. Understand the fundamentals of computer programming and the various programming languages.
2. Learn how to write efficient and effective code.
3. Develop an understanding of basic data structures and algorithms.
4. Develop an understanding of software engineering principles.
5. Become familiar with debugging techniques.
6. Develop an understanding of the different software development methodologies.
7. Become familiar with the various application development platforms.
8. Become familiar with the process of software testing and quality assurance.
9. Develop an understanding of the importance of user interface design.
10. Develop an understanding of the importance of security in software development.

1. To enable the students to understand about the fundamental concepts of analytical instruments

1. Analytical instruments are used to measure physical and chemical properties of samples.
2. These instruments are used in a variety of fields, such as chemistry, biology, engineering, and medicine.
3. Common analytical instruments include spectrometers, chromatographs, calorimeters, and mass spectrometers.
4. Analytical instruments are important tools for research, development, and quality control.
5. Analytical instruments measure properties such as composition, structure, and concentration.

6. The accuracy and precision of analytical instruments depend on the type of instrument and the sample being analyzed.
7. Proper operation and maintenance of analytical instruments is essential for accurate results.

2. Analysis of Chloride Content, Hardness, and Alkalinity

1. Chloride content is the amount of chloride ions present in a given sample of water. It is measured in milligrams per liter (mg/L).
2. Hardness is the measure of the amount of calcium and magnesium ions present in a given sample of water. It is measured in milligrams per liter (mg/L).
3. Alkalinity is the measure of the amount of alkaline substances present in a given sample of water. It is measured in milligrams per liter (mg/L).
4. Analysis of chloride content, hardness, and alkalinity is important in determining the quality of water.
5. Chloride content, hardness, and alkalinity can be determined by using titration techniques and chemical tests.
6. The results of the analysis can be used to assess the suitability of the water for various purposes, such as drinking, agriculture, and industrial uses.

Water

1. Water is a colorless, odorless, tasteless liquid that is essential for all known forms of life.
2. Water covers 71% of the Earth's surface and is vital for all known forms of life.
3. Water is a liquid at temperatures between 0 and 100 degrees Celsius.
4. Water is composed of two atoms of hydrogen and one atom of oxygen, and is represented by the chemical formula H_2O .
5. Water has a high surface tension, which makes it an excellent solvent for many substances.
6. Water has a high specific heat capacity, which means it takes a lot of energy to heat it up or cool it down.
7. Water has a very high boiling point, making it a good choice for many industrial processes.
8. Water has a high density, which makes it a good choice for storing energy.
9. Water is essential for all known forms of life, as it is used for many metabolic processes and for maintaining homeostasis.
10. Water is also important for transportation, as it is used to transport goods and people.

To enable the students to understand about the measure of pH, surface tension and viscosity of

- pH is a measure of the acidity or alkalinity of a solution. It is measured on a scale of 0 to 14, with 0 being the most acidic and 14 being the most alkaline.
- Surface tension is a measure of the elasticity of a liquid's surface. It is measured in millinewtons per meter (mN/m).
- Viscosity is a measure of a liquid's resistance to flow. It is measured in centipoise (cP).

A Liquid

A liquid is a type of matter that has a constant volume but takes the shape of its container. It has a distinct surface and is able to flow. Liquids are made up of particles that are in constant motion and have a low level of attraction to each other.

1. Properties of a Liquid
 - Liquids have a fixed volume but take the shape of their container.
 - Liquids have a distinct surface.
 - Liquids are able to flow.
 - Liquids are made up of particles that are in constant motion.
 - Liquids have a low level of attraction to each other.
2. Examples of Liquids
 - Water
 - Alcohol
 - Oil
 - Juice
 - Milk
 - Mercury

4. To enable the students to understand about the preparation of different resins

- Resins are produced from natural sources, such as tree sap, or synthetically, such as epoxy.
- The production of resins involves the extraction of the raw material and the conversion of it into a usable form.
- A variety of processes are used to produce different resins, such as polymerization, condensation, and cross-linking.
- The properties of a resin depend on the type of raw material used, the type of process used to produce it, and the additives used to modify the properties of the resin.
- Resins can be used in a wide range of applications, such as adhesives, coatings, and composites.

Synthesis of Organic Compounds such as Adipic

1. Organic compounds are molecules that contain carbon and hydrogen atoms, and may also contain other elements such as oxygen, nitrogen, sulfur, and phosphorus.
2. Organic compounds can be synthesized from simpler compounds through a variety of reactions.
3. One such reaction is the condensation reaction, which involves the combining of two molecules with the release of a small molecule such as water.
4. Adipic acid is an example of an organic compound that can be synthesized through a condensation reaction.
5. The condensation reaction of adipic acid involves the combination of two molecules of cyclohexanone with the release of one molecule of water.
6. The reaction is catalyzed by an acid such as sulfuric acid, and the resulting product is adipic acid.
7. Adipic acid is a white crystalline solid that is used in the manufacture of nylon and other synthetic fibers.
8. It is also used in the production of polyurethane foam and other plastics.
9. Adipic acid can be synthesized from other organic compounds as well, such as glucose, through a variety of reactions.
10. The synthesis of organic compounds such as adipic acid is an important part of organic chemistry, and understanding the process can help students gain a better understanding of the subject.

Acid and Paracetamol by Conventional and Green Route

- Acids are substances that release hydrogen ions when dissolved in water. They are commonly used in many industries, such as food and beverage, pharmaceutical, and chemical.
- Paracetamol is a synthetic compound used to treat fever and pain. It is commonly found in over-the-counter medications and is available in both tablet and liquid forms.
- Conventional routes of synthesis involve the use of hazardous chemicals and can be expensive.
- Green routes of synthesis are more environmentally friendly and cost-effective. They involve the use of natural products such as plant extracts and enzymes.
- Synthesis of acid and paracetamol using green routes typically involves the use of biocatalysts, such as enzymes and microorganisms. These biocatalysts can be used to catalyze the reaction and reduce the amount of energy and hazardous chemicals required.
- The use of green routes for the synthesis of acid and paracetamol can lead to improved product quality, reduced environmental impact, and cost savings.

List of Experiments

1. The Mpemba effect: This is an experiment which investigates the phenomenon where hot water can freeze faster than cold water.
2. The Bernoulli Principle: This experiment examines how an increase in the speed of a fluid results in a decrease in pressure.
3. The Law of Conservation of Mass: This experiment seeks to demonstrate how mass is conserved in a chemical reaction.
4. Boyle's Law: This experiment investigates the relationship between the pressure and volume of a gas.
5. The Ideal Gas Law: This experiment seeks to demonstrate the relationship between pressure, volume, and temperature of a gas.
6. The Doppler Effect: This experiment investigates how the frequency of a sound wave changes when the source is moving.
7. Archimedes Principle: This experiment examines the relationship between the buoyant force and the weight of an object in a fluid.

1. Calibration of Analytical Equipment and Apparatus

- Calibration is the process of verifying that an instrument is properly measuring a physical quantity.
- It is important to ensure that the instrument is correctly calibrated in order to obtain accurate results.
- Calibration can be done in two ways: by comparison to a standard or using a calibration curve.
- When calibrating using a standard, the instrument is compared to a known reference material.
- When calibrating using a calibration curve, a series of measurements are taken at known concentrations and a graph is constructed.
- The instrument is then adjusted until the readings match the graph.
- Calibration should be done regularly to ensure accuracy of results.
- In addition, it is important to document the results of the calibration for future reference.

2. Determination of Hardness of Water Sample by EDTA Method

1. Hardness of water can be determined by titration with EDTA (Ethylenediaminetetraacetic acid).
2. EDTA forms a complex with metal ions in the water sample, making them soluble and thus, reducing the hardness of the sample.

3. The titration of EDTA with the water sample is done using a standardized solution of EDTA, which is usually 0.01 M.
4. The titration is performed until the endpoint is reached, which is usually indicated by a color change.
5. The amount of EDTA used in the titration is a measure of the hardness of the water sample.
6. The hardness of the water sample can be calculated using the following formula:

Hardness (in ppm) = (volume of EDTA used (in mL)) x (concentration of EDTA (in M)) x (equivalent weight of calcium carbonate (in g/mol)) x (1000)

3. Determination of Alkalinity of Water Sample

1. Alkalinity is the measure of the capacity of water to neutralize an acid.
2. Alkalinity is an important parameter in water quality and is used to indicate the amount of dissolved minerals in the water.
3. Alkalinity is usually expressed in milligrams per liter (mg/L) of calcium carbonate (CaCO_3).
4. The alkalinity of water can be determined using a titration method.
5. The titration method involves adding a known amount of acid to the water sample and then titrating it with a base such as sodium hydroxide (NaOH) until the sample reaches a pH of 8.3.
6. The amount of acid required to reach this pH is then used to calculate the alkalinity of the sample.
7. The alkalinity of the sample can also be determined by using a colorimetric method.
8. This method involves adding a reagent to the water sample and then measuring the absorbance of the solution at a particular wavelength.
9. The absorbance is then used to calculate the alkalinity of the sample.

4. Determination of pH by titrimetric method

- Titrimetric methods are used to determine the pH of a solution.
- The method involves titrating a solution with a known concentration of acid or base until the endpoint is reached.
- The endpoint is determined by a color change or by using a pH indicator.
- The amount of acid or base used to reach the endpoint is then used to calculate the pH of the solution.
- Titrimetric methods are accurate and can be used in a variety of situations, including the determination of the pH of a buffer solution.

5. Determination of Surface Tension of a Given Liquid

1. Surface tension is a property of a liquid surface that allows it to resist an external force.

2. Surface tension is caused by the cohesive forces between molecules of the same liquid.
3. Surface tension can be measured by the force needed to break the surface of a liquid.
4. A common method of measuring surface tension is the Du Noüy ring method, which uses a ring suspended from a wire to measure the force needed to break the surface of a liquid.
5. The surface tension of a liquid can also be determined by measuring the force needed to move a needle through the surface of the liquid.
6. The surface tension of a given liquid can be used to calculate the surface energy of the liquid, which is a measure of the energy required to break the surface of the liquid.

6. Determination of Viscosity of a given liquid by viscometer

1. Viscometers are instruments used to measure the viscosity of a liquid.
2. The viscosity of a liquid can be determined by measuring the time it takes for a given volume of the liquid to flow through a capillary tube.
3. The viscosity of a liquid is usually expressed in terms of centipoise, which is a measure of the liquid's resistance to flow.
4. To determine the viscosity of a given liquid using a viscometer, a sample of the liquid is placed in the viscometer and the time it takes for the liquid to flow through the capillary tube is measured.
5. The viscosity of the liquid can then be calculated using the equation: $\text{Viscosity} = \text{Time/Volume}$.
6. The viscosity of a liquid can also be determined using a rotational viscometer, which measures the torque required to rotate a spindle immersed in the liquid.
7. The viscosity of a liquid can also be determined using a falling ball viscometer, which measures the time it takes for a ball to fall through a given volume of the liquid.
8. The viscosity of a liquid can also be determined using a bubble viscometer, which measures the time it takes for a bubble to rise through a given volume of the liquid.

7. Determination of the Strength of Ferrous Ammonium Sulfate Using External Indicator

- Ferrous ammonium sulfate (FAS) is a compound composed of iron and ammonium sulfate.
- It is commonly used as a reducing agent in organic chemistry.
- The strength of FAS can be determined using an external indicator.
- The indicator used for this purpose is potassium permanganate (KMnO_4).
- The KMnO_4 is added to a solution of FAS, and the color of the solution is observed.

- If the solution turns pink, it indicates that FAS is present, and the strength of FAS can be determined by the intensity of the pink color.
- The intensity of the pink color is directly proportional to the strength of FAS.
- The strength of FAS can also be determined by titration with a standard solution of KMnO_4 .

8. Determination of the Strength of Potassium Dichromate Using Internal Indicator

1. Potassium dichromate is an inorganic compound with the formula $\text{K}_2\text{Cr}_2\text{O}_7$.
2. It is a strong oxidizing agent, used in a variety of applications such as in the manufacture of dyes, leather tanning, and in analytical chemistry.
3. The strength of Potassium dichromate can be determined by using an internal indicator.
4. The internal indicator used for this purpose is a solution of methyl orange in 0.1 M HCl .
5. The sample of Potassium dichromate is titrated with the internal indicator solution until the end point is reached.
6. The end point is determined by the change in colour of the indicator solution from orange to yellow.
7. The strength of the Potassium dichromate solution is calculated using the following equation:

Strength of Potassium dichromate (M) = (Volume of internal indicator used (mL)) / (Volume of Potassium dichromate sample (mL))

9. Determination of Available Chlorine in Bleaching Powder

1. Bleaching powder is a white, odorless powder that contains calcium hypochlorite and is used as a disinfectant and bleaching agent.
2. Available chlorine is the amount of chlorine in the bleaching powder that is available for disinfecting or bleaching.
3. The available chlorine can be measured by titration with sodium thiosulfate.
4. To determine the available chlorine, a sample of the bleaching powder is dissolved in water and then titrated with a standardized sodium thiosulfate solution.
5. The amount of sodium thiosulfate used in the titration is proportional to the amount of available chlorine present in the sample.
6. The end point of the titration is determined by the addition of a few drops of starch solution, which causes the solution to turn blue.

7. The amount of available chlorine in the sample is calculated based on the amount of sodium thiosulfate used in the titration.

10. Determination of Chloride Content in Water Sample

1. Chloride is an anion found in many water sources, including natural surface and groundwater, and is commonly used to measure water quality.
2. The most common method for determining chloride content in water samples is the titration method. This method involves adding a known amount of a base, such as sodium hydroxide, to the sample and then titrating it with a standard solution of hydrochloric acid.
3. The amount of hydrochloric acid needed to titrate the sample to a neutral pH indicates the concentration of chloride ions in the sample.
4. The chloride content of water can be used to measure its quality, as high levels of chloride can indicate the presence of pollutants such as sewage or industrial waste.
5. It is important to note that the accuracy of the titration method is affected by the presence of other ions in the sample, such as sodium and calcium, as these can interfere with the titration process.

11. Preparation of Phenol formaldehyde (PF) Resin

1. Phenol formaldehyde (PF) resins are synthetic polymers created by the polymerization of phenol and formaldehyde.
2. The polymerization reaction is typically acid-catalyzed and is usually carried out in an aqueous solution.
3. The resulting polymer is a thermosetting polymer that is used in a variety of applications, from adhesives to coatings.
4. The polymerization reaction can be carried out in a variety of ways, including the use of catalysts, solvents, and initiators.
5. The components of the reaction mixture must be carefully selected and balanced to ensure that the desired properties of the finished product are achieved.
6. The reaction can be carried out at a variety of temperatures and pressures, depending on the desired properties of the finished product.
7. The reaction can be carried out in the presence of an initiator, such as an acid or a base, to control the rate of the reaction and the degree of polymerization.

8. The reaction can also be carried out in the presence of a solvent, such as water or alcohol, to control the viscosity of the reaction mixture.
9. Once the reaction is complete, the resulting polymer can be processed in a variety of ways, including extrusion, molding, and coating.
10. The properties of the finished product can be tailored to meet the specific needs of the application.

12. Preparation of Urea Formaldehyde (UF) Resin

1. Urea formaldehyde (UF) resin is a thermosetting polymer produced by the condensation reaction of urea and formaldehyde.
2. UF resin is synthesized by the reaction of urea and formaldehyde in the presence of an acid catalyst at a temperature of 80-90°C.
3. The reaction proceeds in two stages, with the formaldehyde first reacting with the urea to form methylolureas, which then condense to form the final polymer.
4. The degree of polymerization of the UF resin depends on the ratio of urea to formaldehyde used in the reaction, with higher ratios producing higher molecular weight polymers.
5. The UF resin is usually supplied as a powder or granules which can be dissolved in water to form a solution.
6. The UF resin solution is then used as an adhesive or in the manufacture of moulded products such as laminates and particle boards.

13. Preparation of Adipic acid / Paracetamol

- Adipic acid and paracetamol are two important organic compounds used in the manufacture of pharmaceuticals, plastics and dyes.
- Adipic acid is a white, crystalline solid with a melting point of 152°C. It is prepared by the oxidation of cyclohexanol with nitric acid.
- Paracetamol is a white, crystalline solid with a melting point of 169°C. It is prepared by the reaction of phenol with acetic anhydride in the presence of an acid catalyst.
- Both adipic acid and paracetamol are used in the production of various pharmaceuticals, plastics and dyes. Adipic acid is used in the production of nylon, while paracetamol is used in the production of pain relievers.
- Adipic acid and paracetamol can also be used in the synthesis of other organic compounds, such as esters, amides and amines.

14. Determination of Cell Conductance of a Solution

1. Cell conductance is a measure of the ability of a solution to conduct an electrical current.
2. It is important to understand the concept of cell conductance in order to accurately measure and interpret the results of electrical experiments.
3. Cell conductance is dependent on the temperature, concentration, and type of ions present in the solution.
4. To measure the cell conductance of a solution, an electrical current is passed through the solution and the resulting voltage is measured.
5. The cell conductance is then calculated from the voltage and current values.
6. The cell conductance of a solution can be used to determine the concentration of ions present in the solution.
7. It is also used to study the behavior of ions in solutions and to calculate the electrical properties of solutions.

15. Determination of Rate Constant of Hydrolysis of Esters

1. Esters are organic compounds which are formed by the reaction between an acid and an alcohol.
2. The hydrolysis of esters is a reaction in which esters are broken down by reacting with water.
3. The rate of hydrolysis of esters can be determined by measuring the amount of hydrolysis product formed in a given period of time.
4. The rate of hydrolysis is usually expressed in terms of the rate constant, which is the rate of formation of the hydrolysis product per unit time.
5. The rate constant can be determined experimentally by measuring the amount of hydrolysis product formed in a given period of time.
6. The rate constant can also be determined using the Arrhenius equation, which relates the rate constant to the temperature and the activation energy of the reaction.
7. The rate constant can be used to predict the rate of hydrolysis of esters at different temperatures.

16. Element Detection and Identification of Functional Groups in Organic Compounds

- Elements in organic compounds can be identified using a variety of methods, such as spectroscopy, chromatography, and mass spectrometry.
- Functional groups are groups of atoms that are responsible for the chemical

properties of organic compounds.

- Common functional groups include alcohols, carboxylic acids, amines, aldehydes, ketones, esters, and ethers.
- The presence of functional groups can be determined by using a variety of techniques, such as infrared spectroscopy, nuclear magnetic resonance spectroscopy, and mass spectrometry.
- Infrared spectroscopy is a powerful tool for detecting the presence of functional groups by detecting specific absorption peaks.
- Nuclear magnetic resonance spectroscopy can be used to determine the exact structure of a molecule, including the presence of functional groups.
- Mass spectrometry can be used to determine the molecular weight of a compound, which can be used to identify the presence of functional groups.

Study Material

1. Experiment 1:
 - Description of experiment
 - Outcomes
 - Materials needed
2. Experiment 2:
 - Description of experiment
 - Outcomes
 - Materials needed
3. Experiment 3:
 - Description of experiment
 - Outcomes
 - Materials needed
4. Experiment 4:
 - Description of experiment
 - Outcomes
 - Materials needed
5. Experiment 5:
 - Description of experiment
 - Outcomes
 - Materials needed
6. Experiment 6:
 - Description of experiment
 - Outcomes
 - Materials needed
7. Experiment 7:
 - Description of experiment
 - Outcomes
 - Materials needed
8. Experiment 8:
 - Description of experiment

- Outcomes
 - Materials needed
9. Experiment 9:
- Description of experiment
 - Outcomes
 - Materials needed
10. Experiment 10:
- Description of experiment
 - Outcomes
 - Materials needed

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

Course Outcomes:

- Develop an understanding of the foundations of computer science and the principles underlying the design of computer systems.
- Demonstrate the ability to design, implement, and analyze algorithms and data structures.
- Develop an understanding of the principles of operating systems, including process management, memory management, and input/output.
- Demonstrate the ability to program in a high-level language, such as Java, C++, or Python.
- Develop an understanding of the principles of computer networks, including the fundamentals of communication, routing, and switching.
- Demonstrate the ability to design and implement software systems.
- Develop an understanding of the principles of databases, including the fundamentals of data storage and retrieval.
- Demonstrate the ability to design and implement user interfaces.
- Develop an understanding of the principles of artificial intelligence, including the fundamentals of machine learning and natural language processing.
- Explain the fundamentals of computer programming
- Identify and use various programming tools
- Write and execute basic code in a programming language
- Utilize programming libraries, frameworks, and APIs
- Debug code and identify potential errors
- Implement algorithms and data structures

- Design and develop software applications
- Understand and apply object-oriented programming concepts
- Analyze and optimize code for performance and scalability

Course Outcomes Bloom's

1. Understand and explain Bloom's Taxonomy and its application to course outcomes.
2. Analyze course outcomes and determine their level of complexity according to Bloom's Taxonomy.
3. Design course outcomes that align with Bloom's Taxonomy.
4. Create course outcomes that are measurable and assessable according to Bloom's Taxonomy.
5. Develop course outcomes that are applicable to the subject matter and relevant to the course objectives.
6. Evaluate course outcomes and their effectiveness in achieving the course objectives.
7. Synthesize course outcomes to ensure they are comprehensive and meaningful.

Level

A level is a type of measurement used to measure the height or depth of an object. It is used in a variety of fields, such as architecture, engineering, surveying, and construction.

- Levels are typically made of a long, straight, rigid rod with a spirit level attached to one end.
- The spirit level is a device that contains a bubble of air which is used to indicate a horizontal or vertical plane.
- Levels are used to ensure that a surface is level or plumb, or to measure the height of an object.
- They are also used to measure the angle of a slope.
- Levels are available in different sizes and accuracy levels, depending on the application.
- Levels are typically made of metal or plastic, and are often equipped with a handle for easy carrying.

CO-1 Get an Understanding of the Use of Different Analytical Instruments. K3

1. Analytical instruments are tools used to measure, analyze, and interpret physical and chemical properties of substances.

2. Common analytical instruments include spectroscopy, chromatography, mass spectrometry, and microscopy.
3. Spectroscopy is a technique used to measure the intensity of light at various wavelengths. It is used to identify the composition of a substance.
4. Chromatography is a technique used to separate mixtures of substances. It is used to identify and quantify components in a mixture.
5. Mass spectrometry is a technique used to measure the mass-to-charge ratio of ions. It is used to identify and quantify molecules.
6. Microscopy is a technique used to observe small objects. It is used to identify and characterize structures and particles.

CO₂ Measurement of Molecular and System Properties such as Surface Tension

1. Carbon dioxide (CO₂) is a non-polar molecule composed of two oxygen atoms and one carbon atom.
2. The molecular structure of CO₂ is linear, and the molecule has no net dipole moment.
3. The molecular weight of CO₂ is 44.01 g/mol and its molar mass is 0.044 kg/mol.
4. The boiling point of CO₂ is -78.5°C and its melting point is -56.6°C.
5. The density of CO₂ is 1.98 kg/m³ at 0°C and 1.5 kg/m³ at 25°C.
6. The surface tension of CO₂ is 0.042 N/m at 25°C.
7. The viscosity of CO₂ is 0.0014 Ns/m² at 25°C.
8. The solubility of CO₂ in water is 0.25 g/L at 25°C.

Viscosity, Conductance of Solution, Chloride and Iron Content in Water

1. Viscosity is a measure of a fluid's resistance to flow. It is affected by temperature, pressure, and the concentration of dissolved solids. In water, viscosity is affected by the concentration of dissolved salts, such as chloride, and the presence of suspended particles, such as iron.
2. Conductance is a measure of how easily electricity flows through a solution. It is affected by the concentration of dissolved ions, such as chloride and iron. The higher the concentration of dissolved ions, the higher the conductivity.
3. Chloride is an anion (negatively charged ion) that is found in many natural waters. It is an essential nutrient for plants and animals, but can be toxic

in high concentrations.

4. Iron is a micronutrient found in many natural waters. It is essential for healthy plant growth, but can be toxic in high concentrations. Iron can also cause discoloration and staining of surfaces and fabrics.

K3

1. K3 is a type of computer programming language used to create applications and software.
2. It is a strongly typed language, meaning that variables must be declared with a specific type before they can be used.
3. K3 has a syntax similar to other languages such as Java and C++.
4. K3 is an object-oriented language, meaning that objects can be created and manipulated.
5. K3 programs are compiled into bytecode which is then executed by a virtual machine.
6. K3 supports the development of distributed applications, meaning that multiple computers can run the same program.
7. K3 also supports asynchronous programming, which allows for the execution of multiple tasks at the same time.
8. K3 is used for a variety of applications, including web development, mobile development, and game development.

CO-3 Measure the Hardness and Alkalinity of the Water. K3

1. Hardness of water is the measure of the amount of dissolved minerals, primarily calcium and magnesium, present in the water.
2. Alkalinity of water is the measure of the ability of the water to neutralize acids, and is typically expressed as the concentration of calcium carbonate (CaCO_3) in the water.
3. Hardness can be measured using a variety of methods, including titration, atomic absorption spectroscopy, and colorimetry.
4. Alkalinity can be measured using a variety of methods, including titration, ion chromatography, and colorimetry.
5. Hardness and alkalinity are important parameters to consider when evaluating the suitability of water for drinking, irrigation, and other uses.

CO-4 Know the fundamental concepts of the preparation of phenol

1. Phenol is an aromatic compound with a hydroxyl group attached to an aromatic ring.
2. It is a common component of many industrial and consumer products, including plastics, paints, detergents, and drugs.
3. Phenol can be prepared from the reaction of benzene with an oxidizing agent such as chlorine, bromine, or nitric acid.
4. The reaction of benzene with an oxidizing agent produces an intermediate product, a phenol-chloride or phenol-bromide.
5. The phenol-chloride or phenol-bromide can then be hydrolyzed with a base, such as sodium hydroxide, to produce phenol.
6. Phenol can also be prepared by the oxidation of cresols or by the reaction of benzene with an alkali metal.
7. Phenol can also be prepared by the reaction of benzene with an alcohol in the presence of an acid catalyst.
8. Phenol can also be prepared by the reaction of benzene with an alkyl halide in the presence of an acid catalyst.
9. Phenol can also be prepared by the reaction of an alkyne with an alcohol in the presence of an acid catalyst.
10. Phenol can also be prepared by the reaction of an aldehyde with an alcohol in the presence of an acid catalyst.

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

- Formaldehyde is a colorless and flammable gas at room temperature. It is used in the production of many industrial and consumer products, such as building materials, adhesives, and plastics.
- Urea formaldehyde resin is a thermosetting plastic made from urea and formaldehyde. It is used in the production of molded products, such as furniture and kitchenware, and insulation materials.
- Adipic acid is a white, crystalline organic compound used in the production of nylon and polyurethanes. It is also used as a food preservative and flavoring agent.
- Paracetamol is an analgesic and antipyretic drug used to treat pain and fever. It is available in both oral and intravenous forms.

K3

1. K3 is a term used to describe a set of three-dimensional shapes. These shapes are usually composed of three or more faces, edges, and vertices.

2. K3 shapes can be classified into different types, such as prisms, pyramids, and polyhedra. Each type of K3 shape has its own unique characteristics and properties.
3. One of the most common uses of K3 shapes is in the construction of buildings and other structures. K3 shapes are often used as the basis for building frameworks, such as walls and roofs.
4. K3 shapes can also be used in art and design. Artists often use K3 shapes to create sculptures, paintings, and other works of art.
5. K3 shapes are also used in mathematics and science. They can be used to study the properties of three-dimensional space, such as angles, lengths, and areas.
6. K3 shapes are also used in engineering and manufacturing. They can be used to create parts for machines and other products.

CO-5 Estimate the Rate Constant of Reaction K3

1. The rate of a reaction is the speed at which a chemical reaction takes place.
2. The rate constant, K , is a measure of the rate of a reaction and is usually expressed in terms of the concentration of the reactants over time.
3. The rate constant is calculated by measuring the rate at which the concentration of the reactants changes over time.
4. The rate constant is also related to the activation energy of the reaction, which is the energy required for the reaction to take place.
5. The rate constant can be used to determine the temperature at which a reaction will occur and the rate at which it will occur.
6. The rate constant can also be used to predict the rate of a reaction at different temperatures.

Topic: Solar System

1. The Solar System is made up of the Sun, eight planets and their moons, dwarf planets and other small bodies such as asteroids and comets.
2. The Sun is the center of the Solar System, and all the planets and other bodies orbit around it.
3. The planets in the Solar System are Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune.
4. Each planet has its own unique characteristics, such as size, composition, atmosphere and temperature.
5. The moons of the planets are also varied and interesting, with some having atmospheres and others being covered in ice and dust.
6. Dwarf planets, like Pluto, are smaller than planets but larger than asteroids and comets.
7. Asteroids are small rocky bodies that orbit the Sun, while comets are icy

bodies that have a tail of dust and gas when they approach the Sun.

8. The Solar System is part of a larger system called the Milky Way galaxy, which contains billions of stars and planets.

Topic: The Benefits of Exercise

1. Exercise improves physical health by strengthening the heart, lungs, and muscles.
2. Exercise helps to reduce the risk of developing chronic diseases such as heart disease, diabetes, and some types of cancer.
3. Exercise can help to reduce stress and anxiety, as well as improve mood and overall mental health.
4. Exercise can help to improve sleep quality and duration.
5. Exercise can help to reduce and manage body weight.
6. Exercise can help to improve self-confidence and self-esteem.
7. Exercise can help to increase energy levels and improve overall physical fitness.
8. Exercise can help to improve cognitive function, such as memory and concentration.

Topic: The Benefits of AI Assistants

1. AI assistants can provide 24/7 customer service, providing quick and accurate answers to customer inquiries.
2. AI assistants can process large amounts of data quickly, helping to make decisions quickly and accurately.
3. AI assistants can help automate mundane tasks, freeing up time for more important tasks.
4. AI assistants can help improve customer experience by providing personalized recommendations and services.
5. AI assistants can help reduce costs associated with customer service and other tasks.
6. AI assistants can help increase efficiency by providing automated solutions for complex problems.
7. AI assistants can help provide insights into customer behavior, allowing businesses to better understand their customers.
8. AI assistants can help improve accuracy by providing more accurate data analysis.
9. AI assistants can help improve safety by providing automated safety checks.
10. AI assistants can help reduce human error by providing automated solutions.

The Benefits of Regular Exercise

1. Improves physical health: Regular exercise helps to improve overall physical health by increasing muscle strength, reducing body fat levels, and improving cardiovascular health.
2. Improves mental health: Exercise has been shown to reduce stress, anxiety, and depression. It can also help to improve mood and self-esteem.
3. Increases energy levels: Regular exercise can increase energy levels and help to reduce fatigue.
4. Improves sleep: Exercise can help to improve sleep quality, as well as helping to reduce insomnia.
5. Enhances creativity: Exercise has been shown to increase creativity and problem-solving skills.
6. Improves cognitive function: Exercise has been shown to improve cognitive function, including memory, concentration, and focus.
7. Enhances productivity: Regular exercise can help to increase productivity, as it can help to improve focus and concentration.

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1. The importance of being formal in communication cannot be overstated. It is essential to maintain professionalism, respect, and clarity when communicating with others.
2. Emojis should not be used in formal communication as they can be misinterpreted and can detract from the message being conveyed.
3. External links should not be included in formal communication as they can be distracting and can lead to inaccurate information.
4. Markdown is a formatting language used to write documents in a structured and organized manner. It is commonly used for web content, programming, and documentation.
5. Being formal in communication involves using appropriate language, avoiding slang, and adhering to accepted etiquette.
6. When writing study material, it is important to use clear and concise language, provide accurate information, and include relevant examples to illustrate the point.
7. Writing in points allows for easy comprehension of the material and makes it easier to refer back to specific topics.

Topic:

- 1.
- 2.
- 3.

- 4.
- 5.

Please provide a specific topic for me to write about.

ENGINEERING MATHEMATICS LAB

1. Engineering Mathematics is the application of mathematics to solve problems in engineering and science. It is a core subject in engineering and technology studies.
2. Engineering Mathematics covers topics such as calculus, linear algebra, differential equations, probability, Fourier analysis, and numerical analysis.
3. In an Engineering Mathematics lab, students will learn how to use mathematical techniques and software to solve engineering problems.
4. Students will learn how to apply mathematical principles to solve engineering problems, such as designing a bridge or analyzing the performance of a machine.
5. They will also learn how to use mathematical software to solve complex engineering problems.
6. In addition, students will learn how to interpret and analyze data obtained from experiments and simulations.
7. Finally, they will learn how to present their results in a clear and concise manner.

Course Objectives:

1. Understand the fundamentals of computer science and programming.
2. Develop problem-solving skills and apply them to computer science and programming.
3. Develop an understanding of the underlying principles of computer hardware and software.
4. Develop an understanding of the principles of computer networks, security, and data storage.
5. Develop an understanding of the principles of software engineering and software development.
6. Develop an understanding of the principles of database design and management.
7. Develop an understanding of the principles of artificial intelligence, machine learning, and robotics.
8. Develop an understanding of the principles of user interface design and user experience.

1. To enable the students to understand about the fundamental concepts of analytical instruments

1. An analytical instrument is an instrument used to measure and analyze physical properties of a sample.
2. The types of analytical instruments include spectroscopy, chromatography, microscopy, and mass spectrometry.
3. Spectroscopy is the study of the interaction between matter and electromagnetic radiation. It is used to identify and quantify chemical components in a sample.
4. Chromatography is a technique used to separate mixtures of substances into their individual components.
5. Microscopy is the study of small objects using a microscope. It is used to study the structure and composition of a sample.
6. Mass spectrometry is the analysis of mixtures of molecules based on their mass-to-charge ratio. It is used to identify and quantify chemical components in a sample.
7. Analytical instruments are used to measure and analyze physical properties such as composition, structure, and concentration of a sample.

2. To enable the students to understand about the analysis of chloride content, hardness, alkalinity of water

- Chloride content in water can be determined by titration with silver nitrate.
- Hardness of water is typically determined by titration with EDTA, and is usually expressed in terms of calcium carbonate equivalents.
- Alkalinity of water can be determined by titration with a strong acid such as sulfuric acid, and is usually expressed in terms of calcium carbonate equivalents.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of a liquid

- pH is a measure of the acidity or alkalinity of a liquid, measured on a scale from 0 to 14. A neutral pH is 7, and a liquid with a pH below 7 is considered acidic, while one with a pH above 7 is considered basic or alkaline.
- Surface tension is a measure of the cohesive forces between liquid molecules that cause a liquid to act as if it has an elastic surface. It is measured in units of force per unit length.
- Viscosity is a measure of a liquid's resistance to flow. It is measured in units of force per unit area. Higher viscosity liquids are more resistant to flow than lower viscosity liquids.

4. To enable the students to understand about the preparation of different resins

- Resins are materials derived from plants, animals, or synthetic sources.
- Resins are generally composed of complex mixtures of organic compounds, such as terpenes, and can be used for a variety of purposes.
- Resins can be used as adhesives, coatings, and sealants.
- Natural resins can be produced by plants and animals, while synthetic resins are produced from petroleum-based chemicals.
- Resins can be prepared by a variety of methods, including distillation, fractional distillation, extraction, and polymerization.
- Distillation is the process of heating a liquid to produce vapors that can be condensed back into liquid form.
- Fractional distillation is a process of separating components of a mixture based on their boiling points.
- Extraction is the process of separating a solid from a liquid by dissolving it in a solvent.
- Polymerization is the process of combining two or more molecules to form larger molecules.

5. To enable the students to understand about the synthesis of organic compounds such as adipic acid and paracetamol by conventional and green route

1. Synthesis of organic compounds such as adipic acid and paracetamol is a process of combining two or more components to create a new compound.
2. Conventional synthesis of organic compounds involves the use of chemical reagents and catalysts to produce the desired product.
3. Green synthesis is a more sustainable approach to synthesis, which involves the use of renewable resources, natural catalysts, and mild reaction conditions.
4. Adipic acid is a white crystalline solid that is used as a precursor in the manufacture of nylon and other polymers. It is produced by the oxidation of cyclohexanol with nitric acid.
5. Paracetamol is a commonly used analgesic and antipyretic drug. It is synthesized through the conversion of p-aminophenol to acetaminophen.
6. The synthesis of organic compounds such as adipic acid and paracetamol can be achieved by both conventional and green routes. The choice of route depends on the desired product and the available resources.

LIST OF EXPERIMENTS

1. Boyle's Law: This experiment was first performed by Robert Boyle in 1662. It states that the volume of a gas is inversely proportional to the pressure exerted on it.
2. Charles' Law: This experiment was first performed by Jacques Charles in 1787. It states that the volume of a gas is directly proportional to its

temperature when pressure is held constant.

3. Avogadro's Law: This experiment was first performed by Amedeo Avogadro in 1811. It states that the volume of a gas is directly proportional to the number of moles of the gas present when pressure and temperature are held constant.
4. Gay-Lussac's Law: This experiment was first performed by Joseph Louis Gay-Lussac in 1809. It states that the pressure of a gas is directly proportional to the temperature when volume is held constant.
5. Graham's Law: This experiment was first performed by Thomas Graham in 1848. It states that the rate of effusion of a gas is inversely proportional to the square root of its molecular mass.

1. Calibration of Analytical Equipment and Apparatus

1. Calibration is the process of verifying the accuracy of the measurements taken by a particular instrument or device.
2. The process involves comparing the instrument's readings to a known standard and adjusting the instrument's settings to ensure accuracy.
3. Calibration is important to ensure that measurements taken are accurate and reliable.
4. Analytical equipment and apparatus are used in laboratories and other settings to measure physical and chemical properties of substances.
5. This type of equipment may include pH meters, spectrophotometers, chromatographs, thermometers, and balances.
6. In order to ensure the accuracy of the measurements taken by the equipment, calibration is necessary.
7. The process of calibrating analytical equipment and apparatus involves comparing the instrument's readings to a known standard, and adjusting the instrument's settings to ensure accuracy.
8. Calibration should be done regularly to ensure accuracy and reliability of measurements taken.
9. Proper calibration can also help to reduce errors and improve the accuracy of the results.

2. Determination of Hardness of Water Sample by EDTA Method

1. Hardness of water can be determined by a titration method using EDTA (ethylenediaminetetraacetic acid).
2. EDTA is a chelating agent that binds to the metal ions in water, such as calcium and magnesium, and forms a complex.
3. The amount of EDTA required to bind all the metal ions in the sample is determined by titration.
4. The titration is performed by adding a known amount of EDTA to the sample and then titrating with a standard solution of a metal salt, such as calcium chloride.

5. The amount of EDTA required to bind all the metal ions in the sample is determined by the amount of metal salt required to titrate the EDTA.
6. The hardness of the water sample is then calculated by multiplying the amount of metal salt required to titrate the EDTA by the factor for the metal ion being tested.

3. Determination of Alkalinity of water sample

1. Alkalinity is the measure of the capacity of water to neutralize acids.
2. Alkalinity can be determined by titration with a standard acid solution.
3. The titration is usually done with a pH meter and burette.
4. The end point of the titration is determined by a pH indicator.
5. The alkalinity of the sample is calculated from the volume of acid used and the normality of the acid.

4. Determination of pH by Titrimetric Method

- Titrimetric methods involve the use of a titrant, which is a solution of known concentration, to determine the concentration of an unknown solution.
- In this method, the pH of the unknown solution is determined by titrating it with a standard solution of a base or an acid.
- The endpoint of the titration is determined by a color change, by the use of an indicator, or by the use of a pH meter.
- The pH of the unknown solution can then be calculated from the volume of titrant used and the concentration of the titrant.

5. Determination of surface tension of given liquid

- Surface tension is a property of a liquid that causes it to resist an external force, resulting in a surface that acts like an elastic film.
- Surface tension is caused by the cohesive forces between molecules of the same substance that are in contact with each other.
- The surface tension of a liquid can be determined by measuring the force required to break a curved surface of the liquid. This is usually done using a device known as a tensiometer.
- The surface tension of a given liquid can also be determined using a device known as a capillary tube. This device consists of a tube that is partially filled with the liquid to be tested. The force of the surface tension of the liquid causes the liquid to rise up the tube. The height to which the liquid rises is a measure of the surface tension.
- Other methods of determining the surface tension of a liquid involve measuring the pressure required to break a bubble of the liquid or measuring the force required to draw a line on the surface of the liquid.

6. Determination of Viscosity of a given liquid by viscometer

1. Viscosity is a measure of a liquid's resistance to flow and is an important physical property of a liquid.
2. Viscosity can be measured using a viscometer, which is a device that measures the resistance of a fluid to flow.
3. A viscometer consists of two concentric cylinders, one inside the other, with a small gap between them.
4. The inner cylinder is filled with the liquid to be tested and the outer cylinder is filled with a liquid of known viscosity.
5. The viscometer measures the time taken for the inner cylinder to rotate a certain number of revolutions when both cylinders are rotated.
6. The viscosity of the test liquid is then determined from the time taken for the inner cylinder to rotate the specified number of revolutions.
7. The viscosity of the liquid can be calculated by dividing the time taken by the number of revolutions.
8. The viscosity of the liquid can also be determined by measuring the force required to move the inner cylinder through the liquid.

7. Determination of the Strength of Ferrous Ammonium Sulfate Using External Indicator

1. Ferrous ammonium sulfate (FAS) is a salt of ferrous iron and ammonium sulfate.
2. FAS can be used as a reagent in various chemical reactions, such as in the determination of the strength of a solution.
3. An external indicator is a chemical compound that changes color when it comes into contact with another compound.
4. In the determination of the strength of FAS, an external indicator is used to detect the presence of FAS in a solution.
5. The indicator is added to the solution and the resulting color change indicates the strength of the FAS solution.
6. The color change of the indicator is compared to a color chart to determine the strength of the FAS solution.
7. The strength of the FAS solution can then be used to determine the amount of FAS needed for a particular reaction.

8. Determination of the Strength of Potassium Dichromate Using Internal Indicator

1. Potassium dichromate is a strong oxidizing agent and is often used as an internal indicator in titrations.
2. To determine the strength of potassium dichromate, a titration is performed using a standard solution of sodium thiosulphate.
3. The titration is conducted by adding a known volume of sodium thiosulphate to a measured amount of potassium dichromate.

4. The endpoint of the titration is determined by observing the colour change of the potassium dichromate solution.
5. The colour change occurs when the amount of sodium thiosulphate added is equal to the amount of potassium dichromate initially added.
6. The strength of the potassium dichromate solution can then be calculated using the following equation:

$$\text{Strength (g/L)} = \frac{\text{Volume of sodium thiosulphate used (mL)} \times \text{Molarity of sodium thiosulphate (mol/L)}}{\text{Weight (g) of potassium dichromate used}}$$

9. Determination of available chlorine in bleaching powder

1. Bleaching powder is a compound containing chlorine, which is used as a bleaching agent in the textile industry.
2. The available chlorine in bleaching powder can be determined by titration with standard sodium thiosulfate solution.
3. The reaction between the bleaching powder and sodium thiosulfate solution is as follows: $\text{Cl}_2 + 2\text{Na}_2\text{S}_2\text{O}_3 \rightarrow 2\text{NaCl} + \text{Na}_2\text{S}_4\text{O}_6$
4. The end point of the titration is determined by adding a few drops of starch indicator solution to the titrated solution.
5. The end point is marked by the appearance of a blue colour in the titrated solution.
6. The available chlorine in the bleaching powder is calculated from the volume of sodium thiosulfate solution consumed.

10. Determination of Chloride Content in Water Sample

1. Chloride is an anion found in many salts, such as sodium chloride, and is an important constituent of seawater.
2. Chloride can be measured in water samples using a variety of methods, including titration and ion chromatography.
3. The most commonly used method for determining chloride content in water samples is the Mohr titration.
4. The Mohr titration involves the addition of silver nitrate to a water sample, which produces a silver chloride precipitate.
5. The amount of silver chloride precipitate is then titrated with a standard sodium thiosulfate solution to determine the amount of chloride present in the sample.
6. The amount of chloride present in the sample is then calculated using the following equation:

$$\text{Chloride (mg/L)} = \frac{\text{Titration Volume (mL)} \times \text{Normality of Sodium Thiosulfate (N)}}{\text{Sample Volume (mL)}}$$

7. The result of the titration is then compared to the standard chloride concentration for the water sample.
8. The standard chloride concentration for water samples is typically between 250 and 500 mg/L.
9. If the chloride concentration is higher than the standard, it may indicate contamination from industrial or agricultural sources.
10. It is important to test water samples regularly to ensure that the chloride concentration is within the standard range.

11. Preparation of Phenol formaldehyde (PF) Resin

1. Phenol formaldehyde (PF) resin is a thermosetting polymer that is produced by the reaction of phenol and formaldehyde in the presence of an acid catalyst.
2. The reaction is typically carried out at a temperature of 100-130°C and a pressure of 0.5-2.5 MPa.
3. The reaction is exothermic and the heat of reaction is usually sufficient to maintain the reaction temperature.
4. The reaction is usually complete within 2-3 hours, but the reaction time can be extended to 8-10 hours for higher molecular weight resins.
5. The viscosity of the resin increases with increasing molecular weight, and the molecular weight can be controlled by adding an inhibitor such as paraformaldehyde or an amine.
6. The resin can be further modified by adding other monomers such as urea or melamine to increase the water resistance or flame retardancy of the resin.
7. The resin can also be modified by adding fillers such as silica or alumina to reduce the cost and improve the physical properties of the resin.
8. The resin can be used to produce a wide range of products such as laminates, adhesives, coatings, and mouldings.

12. Preparation of Urea formaldehyde (UF) resin

1. Urea formaldehyde (UF) resin is a thermosetting resin that is produced when urea and formaldehyde are reacted together.
2. UF resin is typically used in the production of adhesives, coatings, and moldings.
3. The reaction between urea and formaldehyde is catalyzed by an acid or a base, such as sulfuric acid or sodium hydroxide.
4. The reaction is exothermic, meaning that it produces heat as it progresses.
5. The reaction is typically carried out in the presence of a solvent, such as water or alcohol.
6. The reaction is typically conducted at temperatures between 60-90°C.
7. The reaction typically takes several hours to complete, depending on the desired degree of polymerization.

8. The degree of polymerization can be adjusted by varying the concentration of urea, formaldehyde, and catalyst.
9. The end product of the reaction is a white, semi-solid material that is insoluble in water.
10. UF resins are typically used in the production of adhesives, coatings, and moldings due to their excellent strength, hardness, and durability.

13. Preparation of Adipic acid / Paracetamol

1. Adipic acid is a white crystalline solid with a molecular formula of $C_6H_{10}O_4$. It is an important industrial chemical used in the manufacture of nylon, polyurethane, and other synthetic materials.
2. Paracetamol is a widely used analgesic and antipyretic drug. It is used to treat pain and fever.
3. The preparation of adipic acid involves the oxidation of cyclohexanol with nitric acid. The reaction is carried out in a sulfuric acid medium.
4. The preparation of paracetamol involves the reaction of phenol with acetic anhydride in the presence of an acid catalyst. The product is then purified by recrystallization.
5. Both adipic acid and paracetamol are important chemicals with a wide range of applications. Adipic acid is used in the manufacture of nylon and other synthetic materials, while paracetamol is used as an analgesic and antipyretic drug.

14. Determination of Cell Conductance of a Solution

1. Cell conductance is a measure of the ability of a solution to conduct an electrical current.
2. It is defined as the ratio of the current to the voltage applied across the solution.
3. Cell conductance can be measured by measuring the current and voltage across a solution.
4. The cell conductance of a solution is affected by the type of ions present in the solution, their concentration, and the temperature of the solution.
5. The cell conductance of a solution can be determined using a conductance meter.
6. The conductance meter measures the current and voltage across the solution and calculates the cell conductance.
7. The cell conductance of a solution can also be determined by measuring the resistance of the solution.
8. The resistance of a solution can be measured using a Wheatstone bridge circuit.
9. The cell conductance of a solution can also be determined by measuring the capacitance of the solution.
10. The capacitance of a solution can be measured using an impedance analyzer.

15. Determination of Rate constant of Hydrolysis of Esters

- Rate constant of hydrolysis of esters is determined by monitoring the disappearance of the ester and the formation of the acid and alcohol products of the reaction.
- The rate constant of hydrolysis is calculated by plotting the logarithm of the concentration of the ester against time.
- The rate of hydrolysis is also affected by the presence of catalysts and other factors such as temperature, solvent, and pH.
- The rate constant of hydrolysis can be used to predict the stability of esters in a particular environment.
- The rate constant of hydrolysis can be determined using a variety of methods such as spectrophotometric, chromatographic, and calorimetric techniques.
- The rate constant of hydrolysis can also be used to develop a kinetic model of the reaction.

16. Element Detection and Identification of Functional Groups in Organic Compounds

1. Elements in organic compounds can be identified using a variety of methods, such as infrared spectroscopy, mass spectrometry, and nuclear magnetic resonance.
2. Functional groups are small clusters of atoms that are responsible for the chemical behaviour of organic compounds. They are identified by their characteristic chemical and physical properties.
3. Infrared spectroscopy is a powerful tool for the detection and identification of functional groups in organic compounds. It uses the absorption of infrared radiation to determine the functional group present.
4. Mass spectrometry is another powerful tool for the detection and identification of functional groups in organic compounds. It uses the fragmentation of molecules to determine the functional group present.
5. Nuclear magnetic resonance (NMR) is a powerful tool for the detection and identification of functional groups in organic compounds. It uses the resonance of nuclei in a magnetic field to determine the functional group present.
6. Chromatography is a powerful tool for the detection and identification of functional groups in organic compounds. It uses the separation of molecules to determine the functional group present.

Topic: Experiments

1. Perform an experiment to determine the boiling point of water.
2. Conduct an experiment to measure the latent heat of vaporization of water.
3. Perform an experiment to determine the specific heat capacity of a material.

4. Conduct an experiment to measure the thermal conductivity of a material.
5. Perform an experiment to measure the viscosity of a liquid.
6. Conduct an experiment to measure the surface tension of a liquid.
7. Perform an experiment to measure the rate of diffusion of a gas.
8. Conduct an experiment to measure the rate of osmosis of a solution.
9. Perform an experiment to measure the rate of chemical reaction of a solution.
10. Conduct an experiment to measure the electrical conductivity of a material.

Course Outcomes:

- Develop an understanding of the fundamentals of computer science and its applications in the world today.
- Gain an understanding of the principles of programming and software development.
- Learn to apply problem-solving techniques to design and implement software solutions.
- Develop the ability to work collaboratively on software projects.
- Develop the ability to analyze and evaluate the effectiveness of software solutions.
- Understand the impact of computing on individuals, organizations, and society.

Upon completion of the course the student should be able to:

- Understand the fundamentals of computer science
- Identify and describe the various components of computer systems
- Explain the principles of algorithmic problem-solving
- Construct and analyze algorithms
- Implement algorithms in a programming language
- Understand the principles of data structures
- Design and implement data structures
- Understand the principles of computer networks
- Understand the principles of operating systems
- Design and implement distributed systems
- Understand the principles of database systems
- Design and implement database systems
- Understand the principles of artificial intelligence
- Design and implement intelligent systems

Course Outcomes Bloom's

1. Understand the basic principles of Bloom's Taxonomy and its application in educational settings.

2. Demonstrate an understanding of the various levels of Bloom's Taxonomy and how they relate to each other.
3. Explain the importance of Bloom's Taxonomy in the development of learning objectives.
4. Utilize Bloom's Taxonomy to create effective learning objectives that are measurable and attainable.
5. Analyze and evaluate student performance using the various levels of Bloom's Taxonomy.
6. Design and implement instruction that is based on Bloom's Taxonomy.
7. Develop assessment instruments that measure student learning outcomes based on Bloom's Taxonomy.

Level

1. Level is a term used to describe a position or status in relation to others.
2. It is often used to describe the rank or grade of a person in a certain field, such as a profession, or in academia.
3. It can also be used to describe a person's social standing or economic class.
4. Levels can be determined in a variety of ways, such as through experience, education, or a combination of both.
5. A person's level can also be determined by the amount of responsibility they have in their job or profession.
6. In some cases, a person's level can be determined by the amount of money they make or the amount of power they have.
7. Levels can also be used to describe the difficulty of a task or activity.
8. For example, a task can be described as being at a beginner, intermediate, or advanced level.
9. Levels can also be used to describe the quality of a product or service.
10. For example, a product may be described as being at a basic, intermediate, or premium level.

CO-1 Get an Understanding of the Use of Different Analytical Instruments. K3

1. Analytical instruments are used to measure and analyze physical and chemical properties of substances.
2. These instruments can measure properties such as mass, temperature, pressure, pH, viscosity, electrical conductivity, and optical properties.
3. They can be used to identify and quantify the components of a sample, as well as to compare the properties of different samples.
4. Some of the most commonly used analytical instruments include spectrophotometers, chromatographs, refractometers, and microscopes.
5. Each instrument has its own set of advantages and disadvantages, so it is important to choose the right instrument for the job.
6. Proper maintenance and calibration of the instrument is also essential for accurate and reliable results.

CO-2 Measure the Molecular / System Properties such as Surface Tension

- Carbon dioxide (CO₂) is an important component of the Earth's atmosphere, and its concentration affects many physical and chemical properties of the atmosphere.
- CO₂ molecules are non-polar and have a linear structure, with two oxygen atoms on either side of the carbon atom.
- The surface tension of CO₂ is determined by the molecules' ability to form hydrogen bonds with other molecules.
- The surface tension of CO₂ is affected by temperature, pressure, and the presence of other molecules in the system.
- At standard temperature and pressure (STP), the surface tension of CO₂ is 32.1 mN/m.
- At higher temperatures, the surface tension of CO₂ decreases, as the molecules become more mobile and more likely to break hydrogen bonds.
- At lower temperatures, the surface tension of CO₂ increases, as the molecules become less mobile and more likely to form hydrogen bonds.
- The surface tension of CO₂ is also affected by the presence of other molecules in the system. For example, the presence of water molecules in the system can increase the surface tension of CO₂.

Viscosity, Conductance of Solution, Chloride and Iron Content in the Water

- Viscosity is a measure of a fluid's resistance to flow. It is determined by measuring the amount of force required to move a given amount of fluid through a given distance.
- Conductance of solution is a measure of the ability of an aqueous solution to conduct electricity. It is determined by measuring the amount of electrical current that is conducted by the solution.
- Chloride is an anion found in many aqueous solutions. It is a major component of seawater and is also found in many industrial and municipal wastewater effluents.
- Iron is an essential nutrient for many organisms. It can be found in many natural waters, but its concentration can vary greatly depending on the source. Iron can also be released into the environment from industrial and agricultural activities.

K3

1. K3 is a computer science field that focuses on the development of software and applications that are designed to help people in their daily lives.
2. K3 is an acronym for Knowledge, Intelligence, and Creativity.
3. K3 applications are designed to be intuitive and user-friendly, making it easier for people to access the information they need.

4. K3 applications are often used to improve user experience and create better user interfaces.
5. K3 applications are used in a wide range of industries, including healthcare, finance, education, and retail.
6. K3 applications can be used to automate processes, provide insights, and make decisions.
7. K3 applications use artificial intelligence and machine learning algorithms to analyze data and make predictions.
8. K3 applications are also used to create virtual assistants, such as chatbots and voice-activated systems.

CO-3 Measure the hardness and alkalinity of the water. K3

1. Hardness of water is a measure of the total amount of dissolved minerals, primarily calcium and magnesium, in the water.
2. Alkalinity is a measure of the capacity of water to resist changes in pH.
3. Hardness is commonly measured in milligrams per liter (mg/L) or parts per million (ppm) of calcium carbonate (CaCO_3).
4. Alkalinity is usually measured in milligrams per liter (mg/L) of calcium carbonate (CaCO_3).
5. Hardness can be determined by titration with a standard EDTA solution.
6. Alkalinity can be determined by titration with a standard acid solution.
7. Hardness of water is important because it affects the taste and can cause scale buildup in pipes and water-using appliances.
8. Alkalinity of water is important because it helps to buffer pH changes and protect aquatic life.

CO-4 Know the fundamental concepts of the preparation of phenol

1. Phenol is an organic compound that is commonly used in the manufacture of plastics, pharmaceuticals, and other products.
2. The most common method of preparing phenol is by the hydrolysis of cumene.
3. Cumene is an alkylbenzene that is produced from the catalytic reforming of petroleum.
4. The hydrolysis of cumene is conducted in an acidic medium, such as sulfuric acid, and results in the formation of phenol and acetone.
5. The reaction is highly exothermic and must be carefully controlled to prevent a runaway reaction.
6. The yield of phenol is usually around 80%, with the remainder being acetone and other byproducts.
7. The purity of the phenol can be increased by recrystallisation, distillation, and other purification processes.

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

- Formaldehyde is a colorless, flammable, strong-smelling chemical used in building materials and many household products. It is used to make products such as plywood, particleboard, and other pressed-wood products; certain insulation materials; glues and adhesives; permanent-press fabrics; paper product coatings; and certain household cleaning products.
- Urea formaldehyde resin is a synthetic resin made by combining urea and formaldehyde. It is used to make adhesives, binders, coatings, laminates, and molding compounds.
- Adipic acid is an organic compound used in the manufacture of nylon, plasticizers, and lubricants. It is also used as a food additive and in the production of some drugs.
- Paracetamol is a widely used analgesic and antipyretic drug. It is used to treat fever, pain, and other symptoms.

K3

1. K3 is a framework for developing and deploying artificial intelligence (AI) applications. It is designed to enable developers to quickly and easily create AI applications that can be deployed on any platform. K3 provides a set of tools and libraries that enable developers to quickly and easily create AI applications. It includes a library of AI algorithms and tools for training and deploying AI models, as well as a set of APIs for integrating AI applications into existing systems. K3 also provides a set of tools for deploying AI applications on multiple platforms, including mobile devices, the web, and cloud services.
2. K3 is designed to be easy to use and to enable developers to quickly and easily create AI applications. It includes a library of AI algorithms and tools for training and deploying AI models, as well as a set of APIs for integrating AI applications into existing systems.
3. K3 provides a set of tools for deploying AI applications on multiple platforms, including mobile devices, the web, and cloud services. It also includes a library of AI algorithms and tools for training and deploying AI models, as well as a set of APIs for integrating AI applications into existing systems.
4. K3 is designed to be secure, reliable, and scalable. It includes a set of security features designed to protect AI applications from malicious attacks, as well as a set of tools for monitoring and managing AI applications.
5. K3 is designed to be extensible and customizable, allowing developers to create AI applications that meet their specific needs. It includes a set of

APIs for integrating AI applications into existing systems, as well as a set of tools for customizing AI applications.

CO-5 Estimate the rate constant of Reaction. K3

1. The rate constant of a reaction is a measure of the speed at which a reaction occurs.
2. It is expressed as the rate of change of concentration of a reactant or product with respect to time.
3. The rate constant can be determined by measuring the rate of change of a reactant or product over time.
4. The rate constant is related to the activation energy of a reaction, and is an important factor in determining the rate of reaction.
5. The rate constant can be calculated from the Arrhenius equation, which relates the rate constant to the activation energy and temperature.
6. The rate constant is also related to the reaction order, which is the number of reactants involved in the reaction.
7. The rate constant can be used to calculate the rate of reaction, which is the rate at which reactants are converted to products.
8. The rate constant can also be used to calculate the equilibrium constant, which is the ratio of the rate of the forward reaction to the rate of the reverse reaction.

Topic:

- 1.
- 2.
- 3.
- 4.
- 5.

Please provide a specific topic for me to write about.

Topic: The Advantages and Disadvantages of Artificial Intelligence

1. Advantages:
 - AI has the potential to automate mundane tasks, freeing up human time and energy for more creative pursuits.
 - AI can process large amounts of data quickly and accurately, providing insights that would not be possible with manual analysis.
 - AI is capable of learning from experience, allowing it to improve its performance over time.

2. Disadvantages:
 - AI can be expensive to develop and maintain, requiring a significant investment of time and resources.
 - AI systems can be vulnerable to hacking or other malicious activities, potentially leading to data breaches or other security issues.
 - AI systems may not be able to replicate human-level creativity, problem-solving, or decision-making abilities.

Topic: The Benefits of Learning a Second Language

1. Improved cognitive abilities: Learning a second language can improve your problem-solving and critical-thinking skills, as well as your memory and attention span.
2. Enhanced communication: Being able to communicate in a second language can open up opportunities to communicate with people from different cultures and backgrounds.
3. Increased job opportunities: Many employers are looking for employees with language skills, as it can make them more competitive in the job market.
4. Increased cultural understanding: Learning a second language can help you understand different cultures and perspectives, which can help you become more open-minded and tolerant.
5. Improved travel experiences: Being able to communicate in a second language can make travelling to foreign countries easier and more enjoyable.

Topic:

- 1.
- 2.
- 3.
- 4.
- 5.

Please provide more information about the topic you would like me to address.

Topic: The Benefits of Exercise

1. Exercise can reduce the risk of developing chronic diseases such as heart disease, diabetes, and certain types of cancer.

2. Exercise can help to reduce stress, anxiety, and depression.
3. Exercise can help to improve sleep quality.
4. Exercise can help to improve muscle strength and endurance.
5. Exercise can help to improve balance and coordination.
6. Exercise can help to improve energy levels.
7. Exercise can help to improve mental focus and clarity.
8. Exercise can help to improve self-confidence and self-esteem.

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1. The purpose of this page is to provide an overview of the topics covered in this section of the study material.
2. This page covers topics such as:
 - The importance of staying informed about current events
 - Understanding the basics of global politics
 - The implications of global economic policies
 - The impact of international trade agreements
3. Staying informed about current events is essential in order to understand the implications of global economic policies and international trade agreements.
4. It is important to understand the basics of global politics, including the roles of different government officials and the power dynamics between countries.
5. Global economic policies can have a significant impact on the global economy, including the prices of commodities, the strength of currencies, and the availability of resources.
6. International trade agreements can have a significant impact on the global economy, including the flow of goods and services, the cost of products, and the ability of countries to access resources.
7. It is important to be aware of the implications of global economic policies and international trade agreements in order to make informed decisions.

Topic:

1. Point 1
2. Point 2
3. Point 3
4. Point 4
5. Point 5

ENGINEERING GRAPHICS LAB KCS

1. Engineering graphics is the graphical representation of engineering objects and their components. It is used to communicate ideas and designs to

other engineers and to the general public.

2. The KCS Engineering Graphics Lab is a computer-aided design (CAD) lab that provides students with the tools and resources to create and visualize engineering designs in a 3D environment.
3. The lab is equipped with a variety of software programs for 3D modeling, rendering, and animation. These programs include AutoCAD, Solidworks, and 3DS Max.
4. The lab also provides access to a variety of 3D printing technologies, such as Fused Deposition Modeling (FDM), Stereolithography (SLA), and Selective Laser Sintering (SLS).
5. In addition to the software and hardware available in the lab, students also have access to a variety of online resources, such as tutorials and webinars, to help them learn the various aspects of engineering graphics.
6. Students are encouraged to use the lab to create their own designs and to experiment with different techniques and technologies. The lab is also open to faculty and staff, who can use the resources for research and teaching purposes.

Course Objectives:

1. Understand the fundamentals of computer programming, including data types, variables, functions, and control flow.
2. Write programs that can solve problems and execute tasks.
3. Learn how to debug programs and identify and fix errors.
4. Understand the principles of object-oriented programming.
5. Use libraries and frameworks to create more powerful programs.
6. Learn best practices for software development, such as writing clean and maintainable code.
7. Utilize version control systems to manage code and collaborate with other developers.

1. To enable the students to understand about the fundamental concepts of analytical instruments

1. Analytical instruments are tools used to measure physical, chemical, and biological properties.
2. These instruments are used to measure the concentration of a substance, the size and shape of particles, the pressure and temperature of a system, and the composition of a material.

3. They are used in a variety of fields, including medicine, engineering, and environmental science.
4. Common types of analytical instruments include chromatographs, spectrometers, microscopes, and mass spectrometers.
5. Each instrument has its own unique set of features and capabilities, and it is important to understand the basics of each instrument before using it.
6. The principles of operation of analytical instruments involve the use of physical, chemical, and biological principles.
7. The use of these principles helps to determine the concentration of a substance, the size and shape of particles, the pressure and temperature of a system, and the composition of a material.
8. The accuracy and precision of the measurements obtained with analytical instruments depend on the quality of the instrument, the calibration of the instrument, and the user's understanding of the instrument.

To enable the students to understand about the analysis of chloride content, hardness, alkalinity

- Chloride content can be determined by titration with silver nitrate using Mohr's method.
- Hardness can be determined by titration with EDTA using a standard calcium solution.
- Alkalinity can be determined by titration with a standard acid solution.

Water

Water is a transparent, tasteless, odorless, and nearly colorless chemical substance, which is the main constituent of Earth's hydrosphere and the fluids of most living organisms. It is vital for all known forms of life, even though it provides no calories or organic nutrients. Its chemical formula is H_2O , meaning that each of its molecules contains one oxygen and two hydrogen atoms, connected by covalent bonds.

Water is the most abundant compound on Earth's surface, covering 70.9% of the planet's surface. It is also the most abundant liquid on Earth. The oceans hold 97% of all Earth's water, while glaciers and polar ice caps account for most of the remaining fresh water.

Water plays an important role in the world economy, as it functions as a solvent for a wide variety of chemical substances and facilitates industrial cooling and transportation. Approximately 70% of freshwater is consumed by agriculture.

Properties of Water

Water has several unique properties that are essential for life on Earth. These properties include:

- **High Specific Heat Capacity:** Water has a high specific heat capacity, which means it can absorb a lot of heat before its temperature changes. This property helps regulate the temperature of the Earth's environment, as it absorbs heat from the Sun and releases it slowly over time.
- **High Heat of Vaporization:** Water has a high heat of vaporization, which means it takes a lot of energy to convert it from a liquid to a gas. This property helps regulate the temperature of the Earth's environment, as it absorbs heat energy from the Sun and releases it slowly over time.
- **Surface Tension:** Water has a high surface tension, which means it is able to form a thin film on the surface of a liquid. This property helps water to move through small spaces, such as the tiny pores in soil.
- **Adhesion and Cohesion:** Water has strong adhesive and cohesive properties, which means it is able to stick to other substances, such as glass and soil particles. This property helps water to move through the environment, as it is able to adhere to surfaces and form droplets.
- **Solvent:** Water is a universal solvent, which means it can dissolve a wide variety of substances. This property is essential for life on Earth, as it allows for the transport of essential nutrients and minerals throughout the environment.

To enable the students to understand about the measure of pH, surface tension and viscosity of

- pH is a measure of acidity or alkalinity of a solution. It is measured on a scale from 0 to 14, with 0 being the most acidic, 7 being neutral and 14 being the most alkaline.
- Surface tension is the ability of a liquid to resist an external force. It is measured in dynes/cm.
- Viscosity is the measure of a fluid's resistance to flow. It is measured in poise or centipoise.

A Liquid

A liquid is a state of matter that has a relatively low viscosity and conforms to the shape of its container. It has a definite volume, but no definite shape. Liquids are sometimes referred to as “fluids”, although this term is often used to refer to all states of matter.

1. Liquids have a definite volume but no definite shape.
2. Liquids are less viscous than solids and gases.

3. Liquids are sometimes referred to as “fluids”.
4. Liquids flow and can take the shape of their container.
5. Liquids have a surface tension which creates a film on the surface of the liquid.
6. Liquids have a relatively low compressibility.
7. Liquids are capable of conveying forces, such as pressure and shear.
8. Liquids can be mixed together to form solutions or suspensions.

4. To enable the students to understand about the preparation of different resins

- Resins are natural or synthetic polymers that are used for a variety of applications, such as coatings, adhesives, and composites.
- The preparation of different resins depends on the type of resin used. For example, epoxy resins are prepared by mixing an epoxide with an amine, while polyester resins are prepared by reacting an alcohol with an acid.
- The curing process is an important part of resin preparation. Curing involves heating or chemically reacting the resin to form a strong and durable material.
- Resins can also be modified with additives to improve their performance. Additives such as fillers, pigments, and catalysts can be added to resins to improve their properties.
- The properties of a resin can be further modified by manipulating the curing process. For example, adjusting the temperature and time of curing can change the properties of a resin.

Synthesis of Organic Compounds such as Adipic

1. Organic compounds are molecules that contain carbon and hydrogen atoms and can also include other elements such as oxygen, nitrogen, and sulfur.
2. Adipic acid is an organic compound with the formula $(\text{CH}_2)_4(\text{COOH})_2$. It is a white, crystalline, odorless solid.
3. Synthesis of adipic acid involves the oxidation of cyclohexanol with nitric acid. This reaction is catalyzed by vanadium (V) oxide, and the product of the reaction is cyclohexanone.
4. Cyclohexanone is then oxidized with nitric acid and catalyzed by vanadium (V) oxide to form adipic acid.
5. Adipic acid can be used to make nylon, polyurethanes, and other polymers. It is also used in the production of food additives, pharmaceuticals, and other industrial chemicals.

Acid and Paracet

Acid and Paracet are two common substances used in a variety of applications.

- Acid is a corrosive substance that is used to break down materials on a molecular level, usually in order to clean or etch them. It can also be used in industrial processes such as metalworking and printing.
- Paracet is an analgesic, or painkiller, that is used to relieve pain and reduce fever. It is available in both over-the-counter and prescription forms.
- Both acid and paracet can be dangerous if used improperly. It is important to follow the instructions on the label and to use protective gear when handling either substance.
- Acid should be stored in a cool, dry place away from heat and direct sunlight. It should also be kept away from children and pets.
- Paracet should be stored in a cool, dry place away from heat, direct sunlight, and moisture. It should also be kept away from children and pets.
- Acid and paracet should never be mixed together, as this can cause a dangerous reaction.
- It is important to dispose of acid and paracet properly. Acid should be disposed of in a chemical waste container, while paracet should be disposed of in a regular trash can.